ATYPICAL LID BUMPS

Lorne Yudcovitch, O.D., M.S., F.A.A.O.

Lecture Acknowledgement and Thanks:

Professor William C. Hefner

Pacific University



Describe this…

- A lesion!

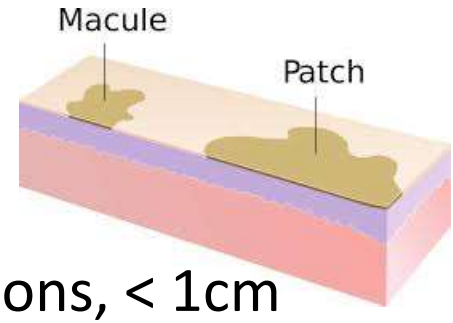


(clinical)

Primary Dermatological Descriptors

AREA

hand & with closed eye u wouldn't notice it



Macule: Circumscribed, flat discoloration , <1cm

Patch: Circumscribed, flat discoloration , >1cm

Papule: Circumscribed, elevated superficial solid lesions, < 1cm

Plaque: Circumscribed, elevated superficial solid lesions, > 1cm

Nodule: Solid lesions with depth above, level or below surface, < 1cm

Tumor: Solid lesions with depth, above, level or below surface, > 1cm

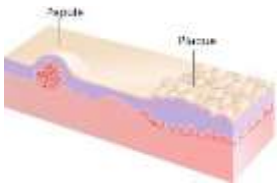
Vesicle: Circumscribed elevations containing serous fluid, < 1cm

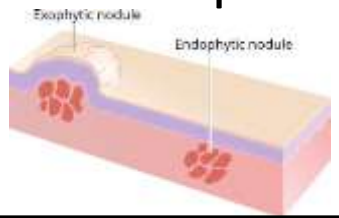
Bigger fluid

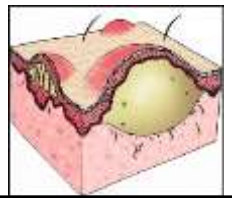
Bulla: Circumscribed elevations containing serous fluid, > 1cm

Petechia: Circumscribed deposits of blood or blood products, < 1cm

Purpura: Circumscribed deposits of blood or blood products, > 1cm



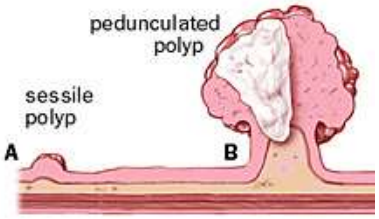


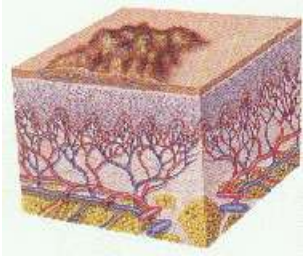


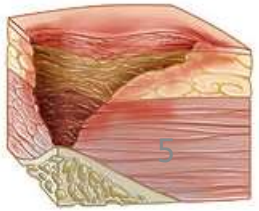


Secondary Dermatological Descriptors

- **Sessile:** A lesion fixed to the skin on a broad base
- **Pedunculated:** A lesion on a stalk
- **Papillomatous:** A lesion exhibiting a surface resembling a cauliflower or artichoke
- **Scales:** Shedding, dead epidermal cells, dry or greasy
- **Umbilicated:** The lesion exhibits a central crater like an umbilicus or belly button
- **Crusts:** Dried masses of skin exudates
- **Ulcer:** Irregularly sized and shaped excavations extending into the dermis







Example

- “A sessile 6mm x 10mm scaly oval nodule with ulcerative umbilicated puprurous center”



Controlled patterns of cell growth

Cancers are uncontrolled cells

1. Hyperplasia
   - Increase in number of cells
2. Metaplasia
   - Change in type of adult cells which is abnormal for that tissue
3. Dysplasia
   - Alteration in size, shape, organization of adult cells

Uncontrolled cell growth

1. Neoplasia
   - Mass of new cells which proliferate without control and serve no useful function
2. Anaplasia
   - loses resemblance to cell of origin
   - AKA tumor or cancer



Tumors can be divided into two general classes:

"nedobri"

benign & malignant

- A malignant tumor can metastasize into lymph nodes & distant organs
- A malignant tumor, if untreated, may kill the patient wherever it occurs. Usually a benign tumor may only cause death if it happens to grow in a vital organ.
- A malignant tumor infiltrates the surrounding tissue. A benign tumor grows by expansion and is usually separated from the surrounding tissue by a capsule.
- When a malignant tumor is excised, it may recur. When a benign tumor is removed, it usually does not recur.
- Generally, a malignant tumor grows more rapidly than a benign tumor.

Nomenclature: by parenchymal cell of origin

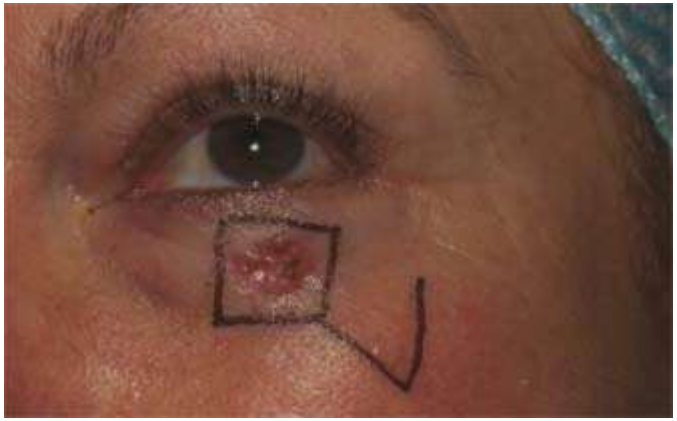
1. **Carcinoma**
   - Epithelial tissue
2. **Sarcoma** (Aids povcano)
   - Tumor of CT (derived from mesoderm), which develops into cartilage and bone (choristoma), fat (lipoma), and muscle (myoma)
3. **Hemangioma**
   - Tumor comprised of blood vessels/vasculature
4. **Lymphoma**
   - Tumor of lymphatic tissue (leukemia – hematopoietic cells affected)
5. **Melanoma** aggresive cancer
   - Tumor of pigmented cells
6. **Hamartoma**
   - Tumor made of tissue normally present there
7. **Blastoma**
   - Tumor derived from embryonic cells (i.e. retinoblastoma)

Tumor Treatment

1. Chemotherapy (Toxic)
2. Cryotherapy (liquid)
3. Radiotherapy
4. Surgery









Neoplastic considerations - Subjective

- Lesion does not act or respond as anticipated
- HX of other skin lesions elsewhere on the body
- HX of other neoplasias or systemic disease
- Pts. with HX of excessive UV skin exposure
- Lesion is not common for patient's age, sex, race, demography
- Older patients
- FHX of skin cancers
- Fair-complexion pts.
- Acute vs. chronic onset & duration
- Rapid or irregular growth patterns
- Pain or irritation associated with suspicious lesion





Neoplastic considerations - Objective

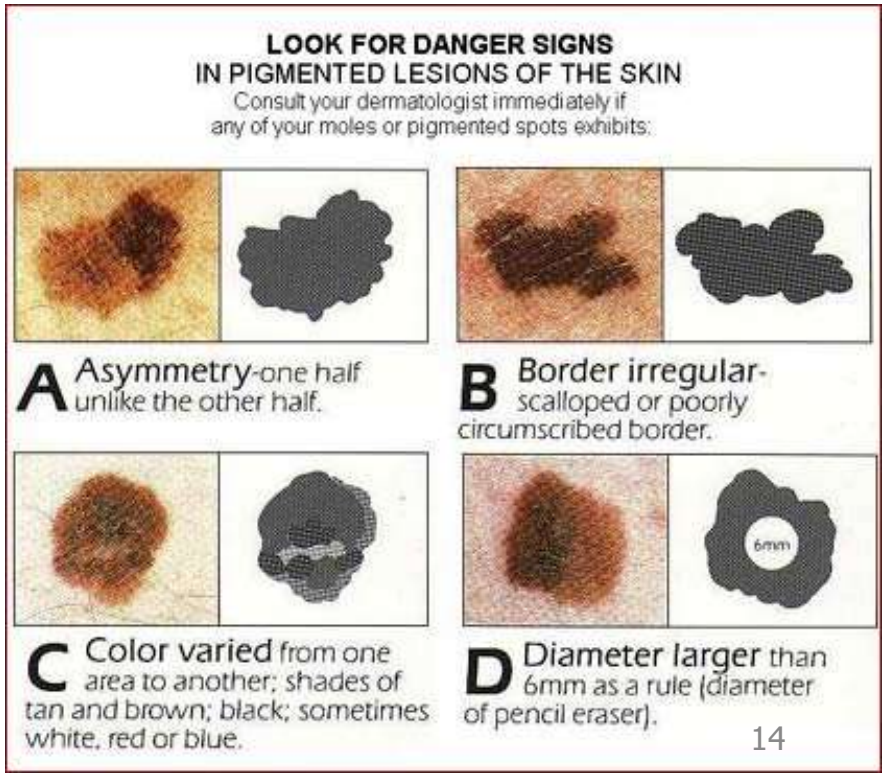


- Quality of the tissue looks irregular
- Bleeding or ulceration of lesion
- Surface integrity & quality questionable
- Changes in lesion not consistent or predictable
- Neovascular patterns around or within lesion
- Recurrent infections or inflammations at site
- Uncharacteristically large lumps or bumps
- Associated tearing or conjunctival hyperemia
- Erosion of the margins or surface of a lesion

Neoplastic Considerations - Assessment / DDX and Plan/TX

- With serious doubt or suspicion regarding any lump or bump of the lid, recommend excision and biopsy
- If referral deferred or refused for any reason, document and monitor closely
  - Photodocument if possible
  - Diagram/measure accurately
- If suspicious, RTC q. 3 mos.
- Rule of thumb :
  - **A**symmetry
  - **B**orders
  - **C**olor
  - **D**iameter
  - (**E**levation) **E**volving

changing during the time



PIGMENTED LESIONS

Ephilis



- = freckle
- Plural: ephilides
- Larger sized melanocytes (normal in number)
- Autosomal dominant inheritance pattern
- Intensified by sun exposure



Ephelides



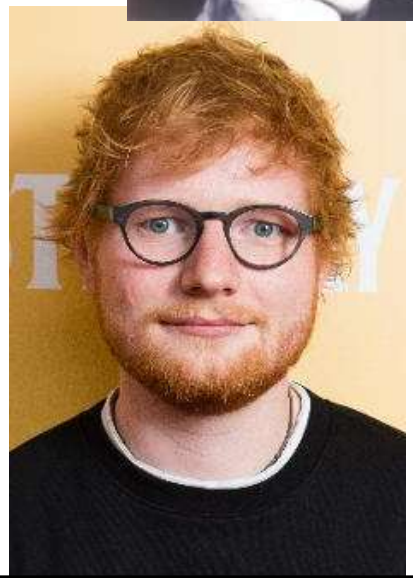


Ephelides…
‘redhead’ connection?













Solar lentigines

- = “liver spots”, age spots
- Occur in response to sunlight
- Persist in the absence of sunlight
- Middle aged to older patients
- Expanding macules (flat patches)
- Normal number of melanocytes



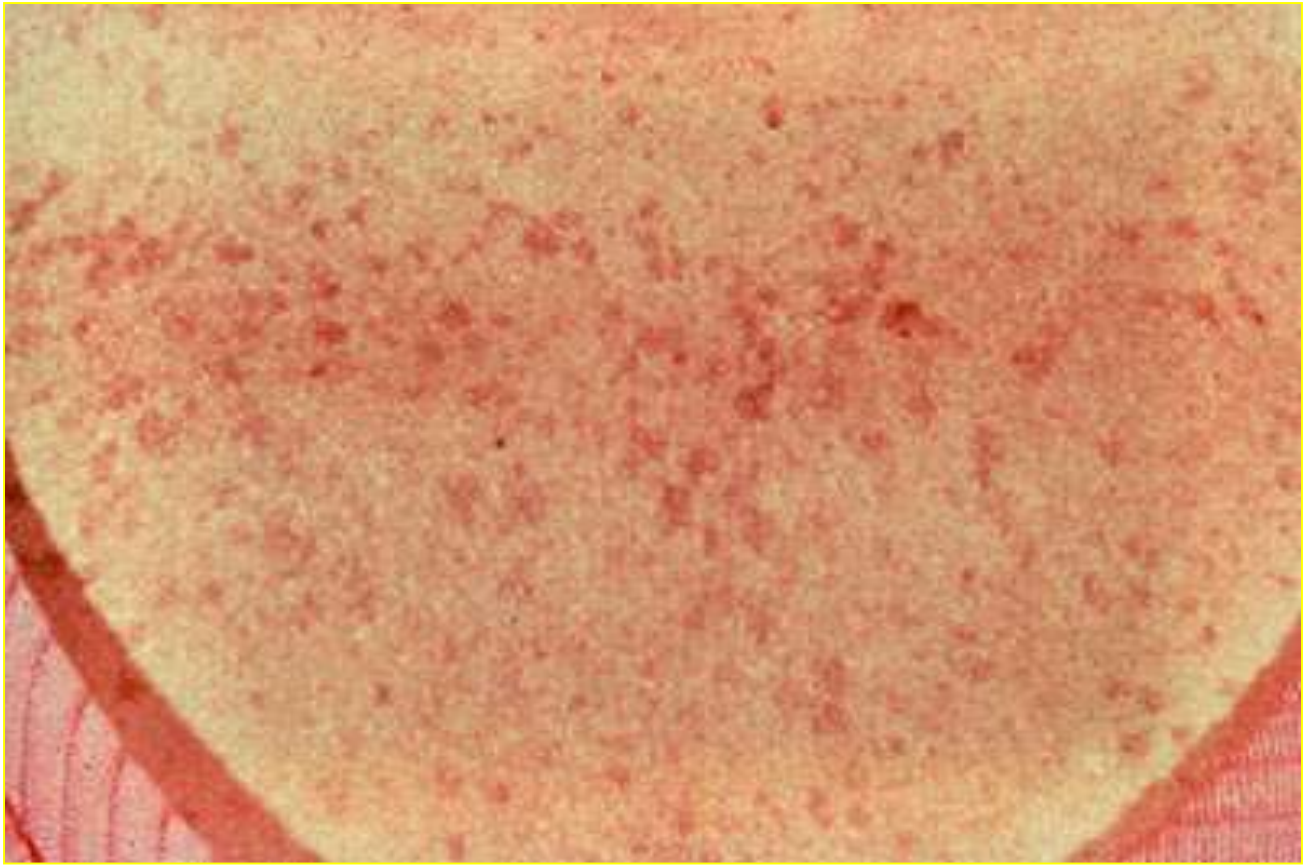




Usually benign



Solar lentigines



Solar lentigines - long time UV exposure





Pre- and post-laser treatment

Simple lentigines



Unlike solar lentigines, simple lentigines:

1. Occur at any age
2. Have no predilection for sun-exposure or lighter skin type
3. Do not darken in response to sunlight

Solar lentigines pre and post laser Tx





- **Mechanism:**
  Pressure waves fragment pigment particles
- **Outcome:**
  Lesions darken or grey over several days to weeks before sloughing off

Nevus

- = mole
- Plural: nevi
- common benign neoplasms or melanocytes
- Subjective:
  - Congenital or early onset
  - Occasional changes in size or pigmentation
  - Cosmetic concern



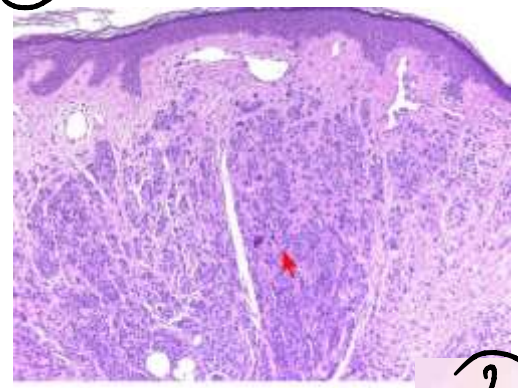


Nevus - Objective (histological)

Several types:

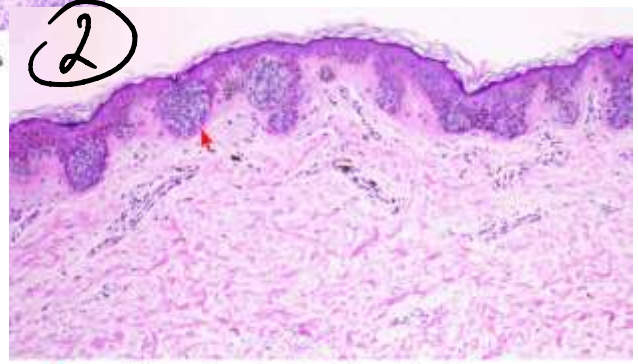
1. Dermal
   - Most common
   - Located in dermis
   - Raised or flat
2. Junctional
   - Dermoepidermal junction
   - Superficial
   - Usually flat
   - May convert to melanoma
3. Compound
   - Both in dermoepidermal junction and dermis
   - Transitional

1



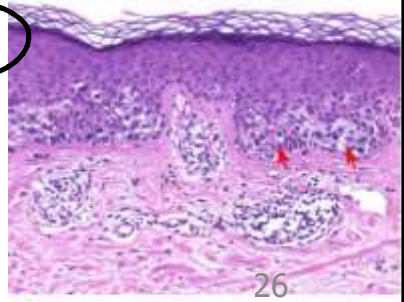
intradermal = only within the dermis

2



"nests" that are junctional (only in epidermis)

3



"nests" along rete ridges that are "bridged" (interconnected along their bases)

Nevus – Dysplastic

- "Fried egg" appearance
- Increased melanoma risk if multiple and with +FHx of melanoma
- Pigmented or amelanotic
- Commonly found at lid margins
- Borders
- Less than 8 - 10 mm diameter
- May increase in size with aging
- Occasionally show hairs growing through surface

lashes/hair are a good sign in nevus



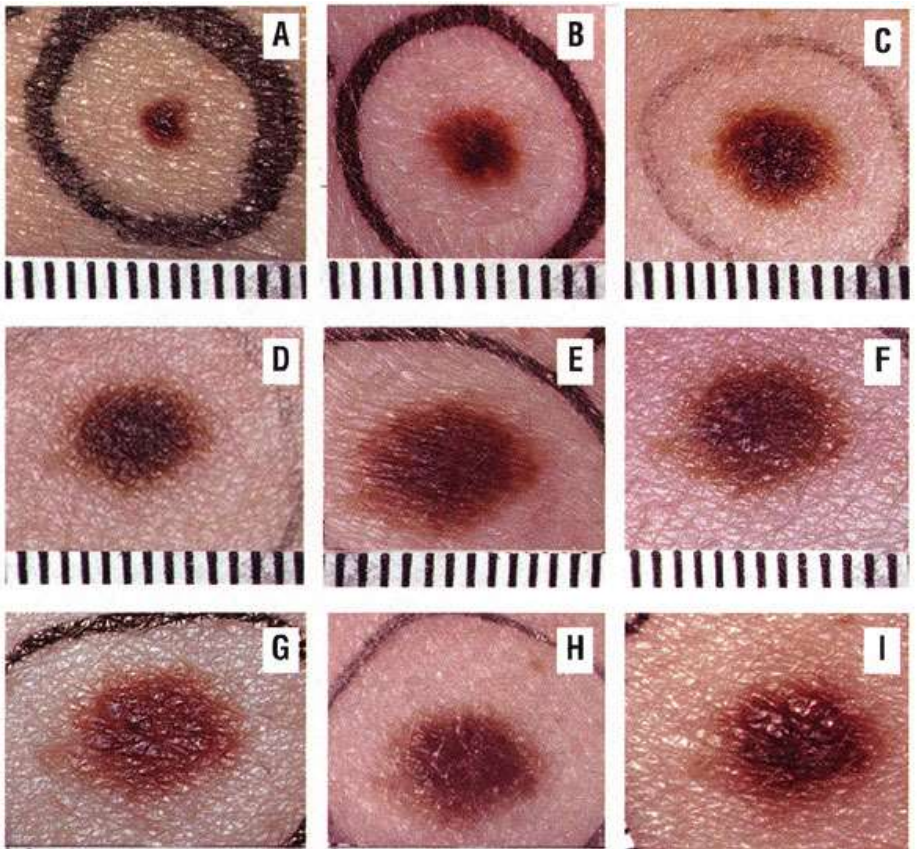
Nevi convert to melanoma have to be careful.

Nevus - Assessment/DDx

- R/O other pigment spots

Plan:

- Photodocument or diagram and measure
- Monitor q. 3 - 6 months if suspicious
- RTC q. year if normal appearance with no change
- Refer for biopsy if changes occur



Dysplastic nevus change over 4 years in 26-year-old white male

Nevus



Dermal Nevus



Dermal Nevus



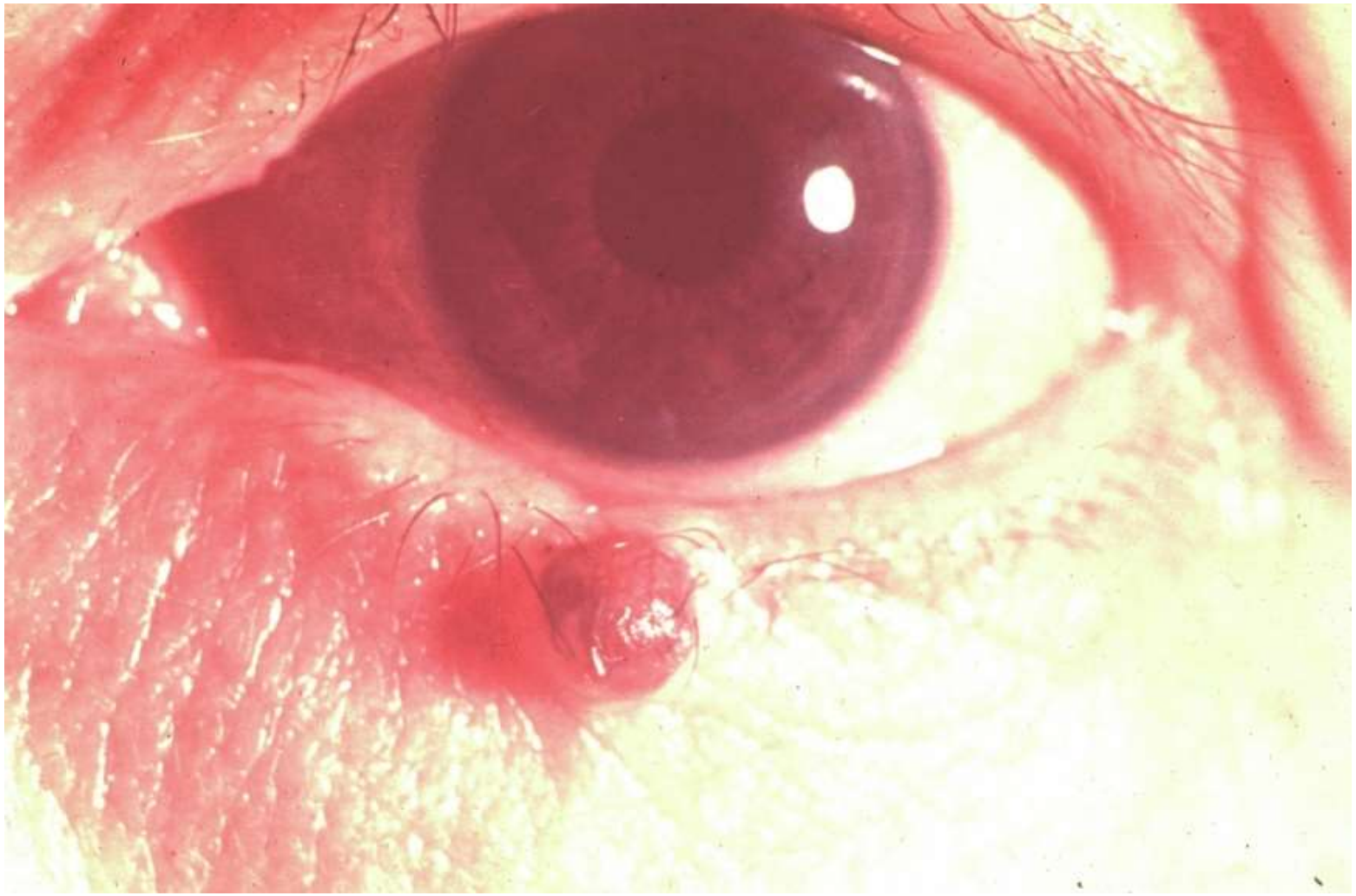
NO melanocites

melanocites



on the junction of skin & conjuctiva

Junctional Nevus

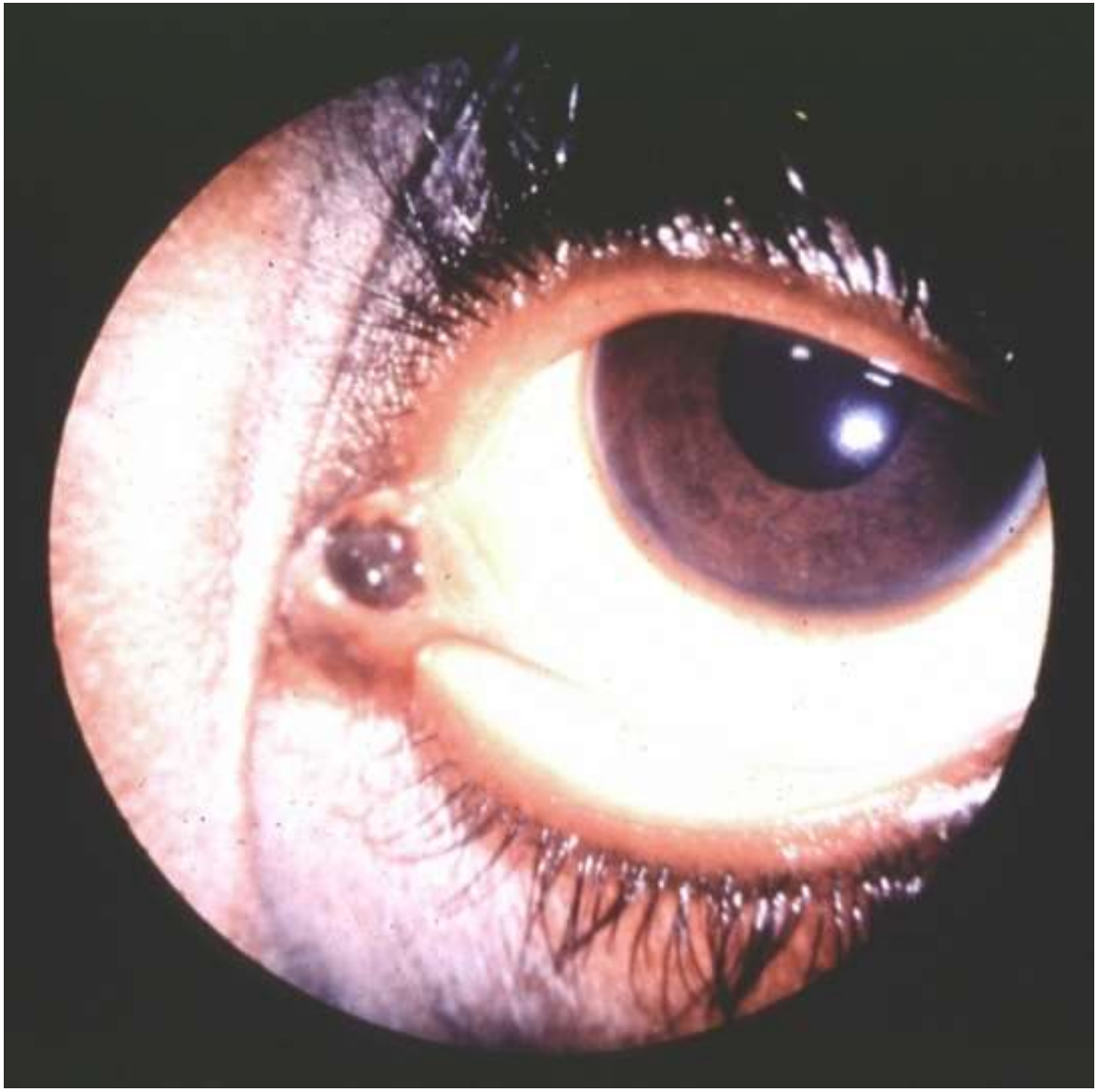


Junctional Nevus





Compound Nevus



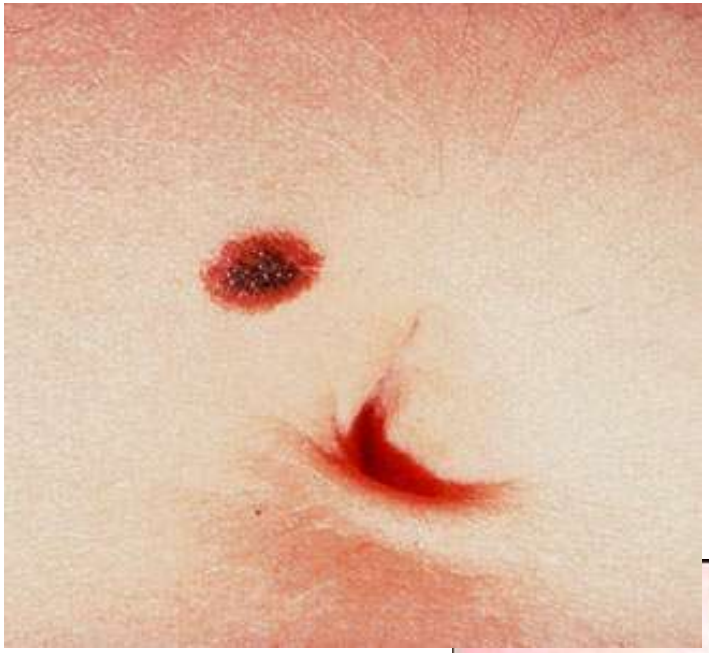
Compound Nevus





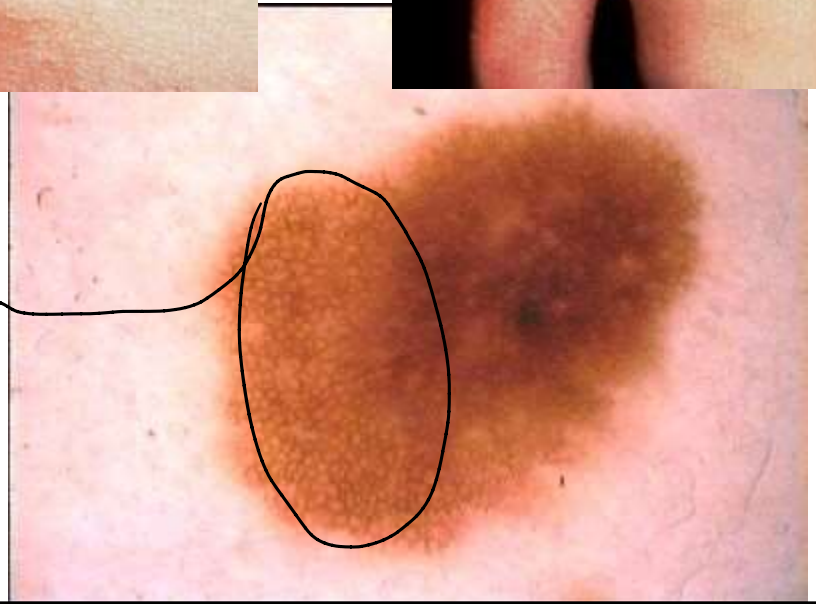
more irregular in shape/colour

Dysplastic Nevus





honeycompound look



MALIGNANT MELANOMA

Malignant Melanoma

TUMOR

life threatening

The most agressive on i spreds quickly

- Malignant tumor derived from melanocytes
- These comprise ~ 3% of primary skin cancers…
- but accounts for 79% of skin cancer related deaths!



Malignant Melanoma - Subjective

- May not directly relate to UV exposure
- Light complexion



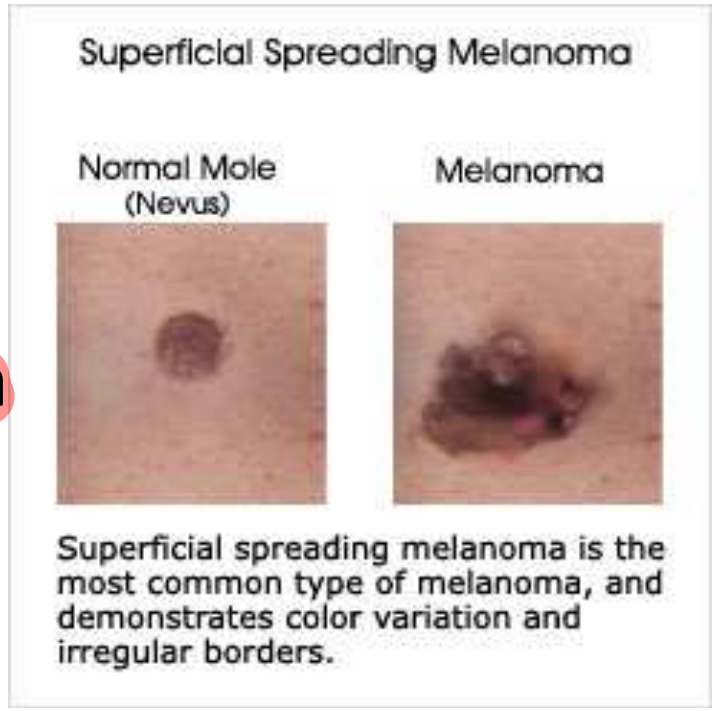


Malignant Melanoma - Objective

- Asymmetric, pigmented lesion with irregular borders
- Variable coloration
- Variable diameter

Types:

1. Superficial spreading melanoma
   - Rapid increase in size
   - Lower incidence in metastases
2. Nodular melanoma
   - Vertical growth with more penetration into tissue
   - Metastasize early





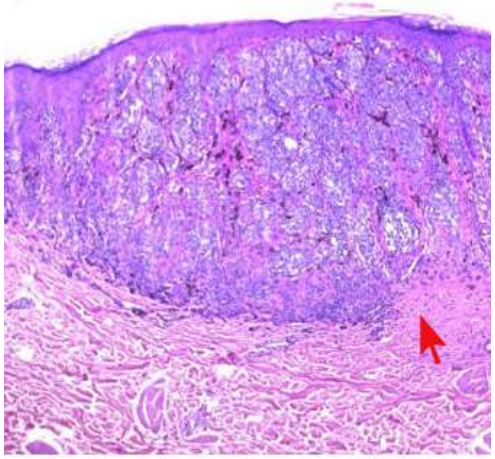
Malignant Melanoma Mortality via Clark's levels and Breslow depth

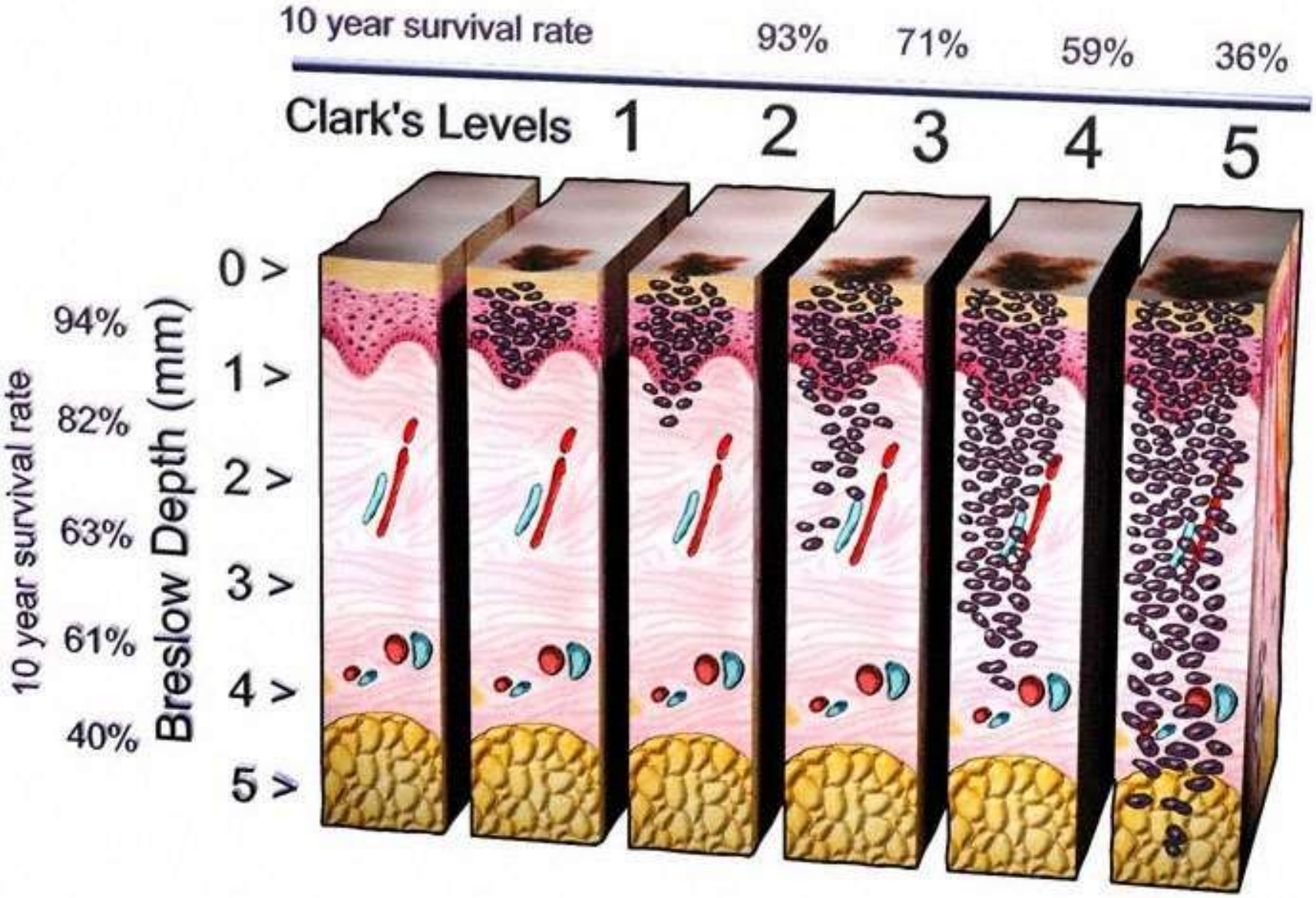
Clark's level – less accurate

- Level I: epidermis; called "in situ" melanoma; 100% cure
- Level II: invasion of the papillary (upper) dermis
- Level III: filling papillary dermis, not reticular (lower) dermis
- Level IV: invasion of reticular dermis → ~50% dead in 10 yr
- Level V: invasion of the deep, subcutaneous tissue

Breslow depth – more accurate

- 0.75 mm deep → 0% metastases
- 0.85 mm deep → 97% survival in 10 years
- 3.6 mm deep → 50% dead in 10 years




10 year survival rate
93%
71%
59%
36%
Clark's Levels
1
2
3
4
5
10 year survival rate
94%
82%
63%
61%
40%
Breslow Depth (mm)
0 >
1 >
2 >
3 >
4 >
5 >

Malignant Melanoma Assessment

- R/O pigmented lesions

Plan:

- Refer for biopsy



Malignant Melanoma



Malignant Melanoma



Malignant Melanoma









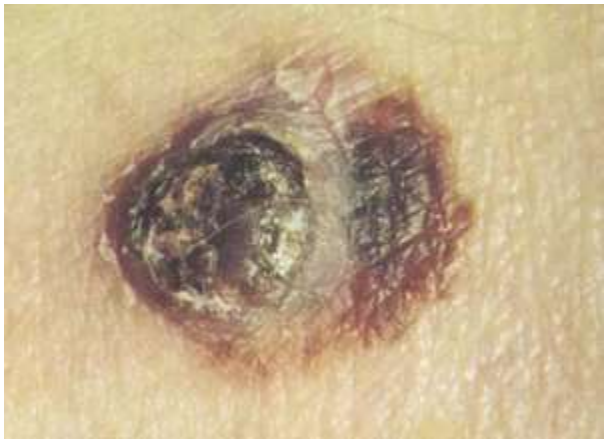
Superficial Spreading Melanoma





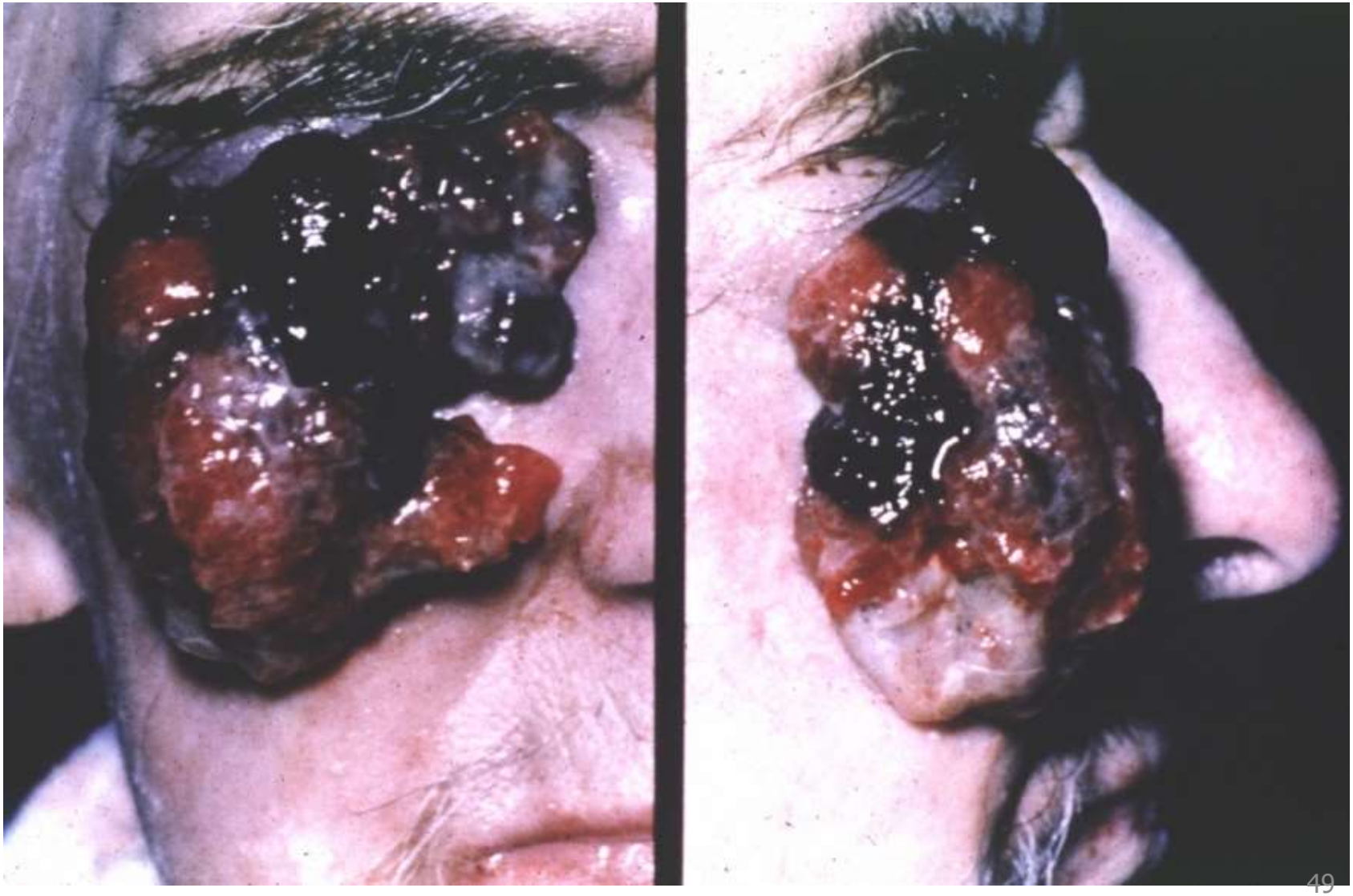

Nodular Melanoma



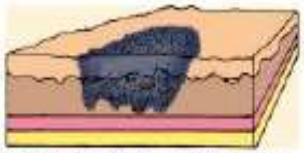


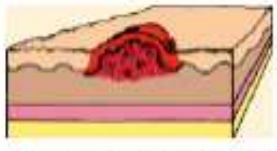


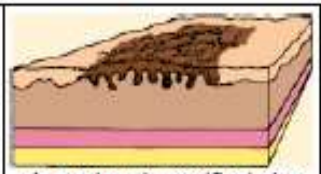
Malignant Melanoma



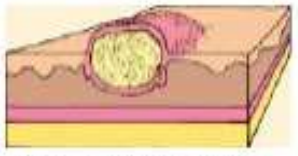
Eyelid Lesions Len V Koh OD & Lorne Yudcovitch OD

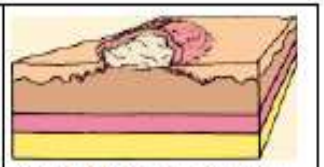

Macule: flat, small, multicolor


Papule: small(<5mm)elevation


Plaque: large(>5mm)flat lesion


Nodule: large solid elevation
Tumor: large nodule


Pustule: purulent elevation

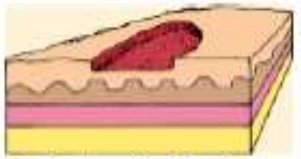

Wheal: edematous skin area

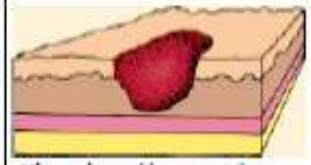

Cyst: enclosed cavity with lining (semisolid/liquid)

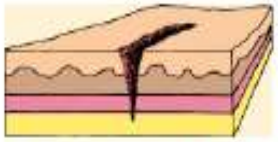

Vesicle: small blister (serum)
Bullae: large vesicle (>5mm)


Scale:dead epidermal cells


Crust: dried exudate 'scab'


Erosion:loss of epidermis


Ulcer: dermal loss, scarring


Fissure: deep dermal skin split


Flat-topped (lichenoid)


Dome-shaped


Pedunculated


Filliform


Verrucous


Umbilicated


Erythema: skin redness


Excoriation


Purpura
Petechiae


Keloid


Lichenification


Sessile squamous papilloma


Pedunculated papilloma


Seborrheic keratosis


Keratoacanthoma


Cutaneous horn


Actinic keratosis


Sebaceous nevus


Basal cell carcinoma


Hemorrhagic BCC


Morpheaform BCC


Squamous cell carcinoma


Sebaceous carcinoma

References: 1. Shields JA; Shields CL. Eyelid, Conjunctival, and Orbital Tumors: An Atlas and Textbook. Philadelphia: Wolters Kluwer Health, 2016. v. 3rd edition
2. Bolognia JL, Schaffer JV, Karynne DO, Ko CJ. Dermatology Essentials. Elsevier Saunders Inc 2014. v. 1st edition

©210521

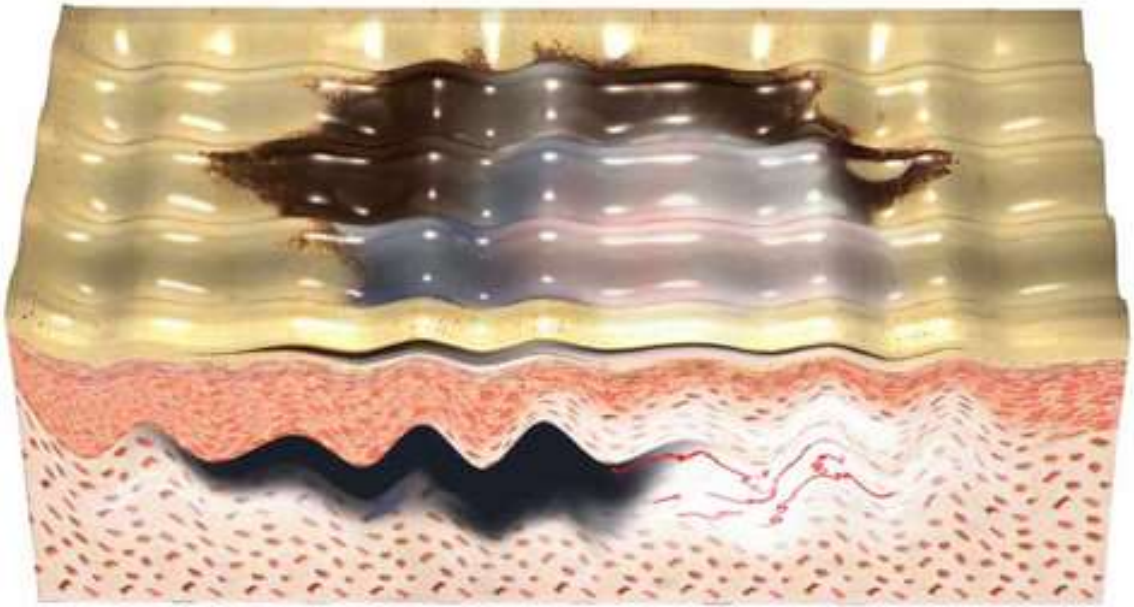
Epiluminescence Microscopy (ELM)

- Typically x10 mag
- Oil medium between instrument and skin
- Allows inspection unobstructed by skin surface reflections
- Makes subsurface structures viewable in vivo
- New versions use polarized light to cancel reflections





Epiluminescence Microscopy (ELM)





- Allows magnified visualization of pigmented skin lesions beyond what is visible in normal clinical exam
- Designed for small equivocal pigmented skin lesions
- Also called: dermatoscopy, dermoscopy, skin surface microscopy, incidence light microscopy

Punch Biopsy





Punch Biopsy



Takes as much as needed, takes little by little as long you don't get the whole lesion out.

Mohs Surgery

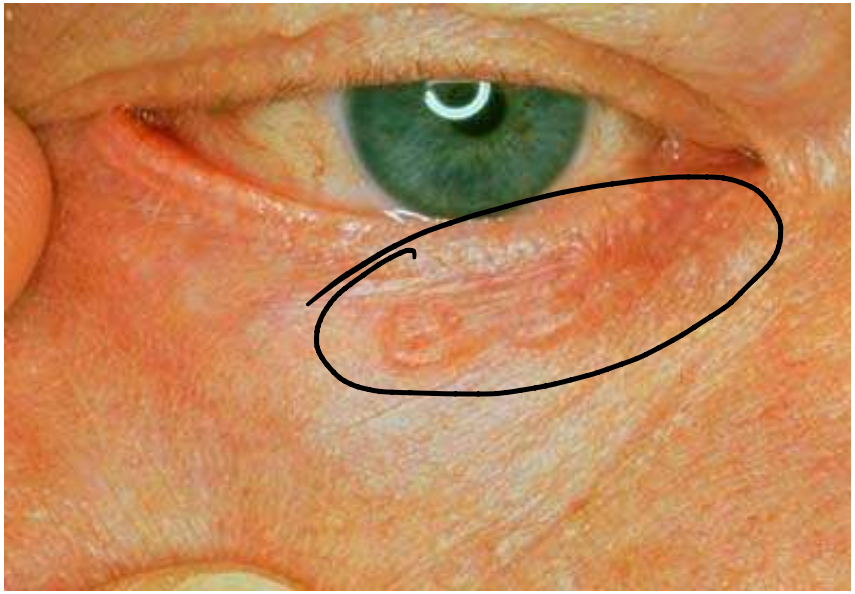


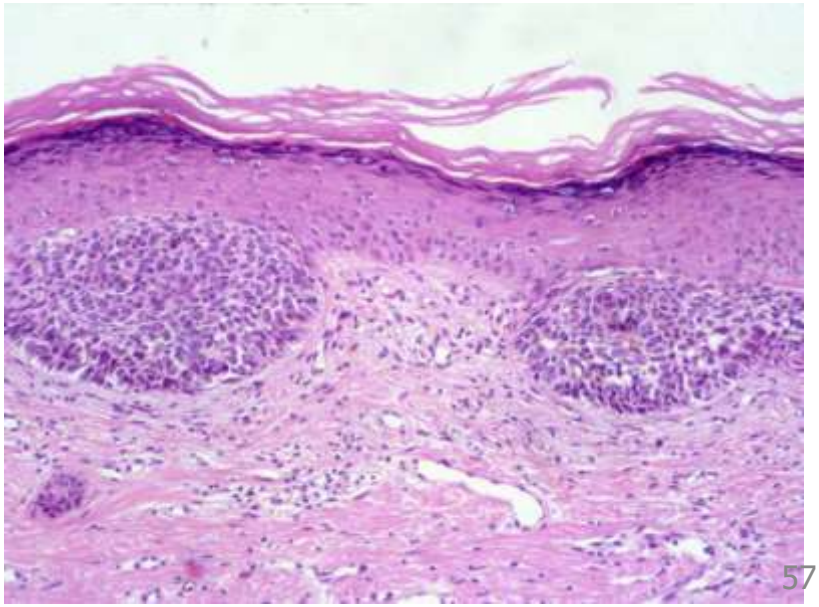
BASAL CELL CARCINOMA

most common cancer of the eye lid

Basal Cell Carcinoma

- Malignant tumor
- MOST COMMON MALIGNANT LID TUMOR
- Etiology – UV exposure connection, elderly, fair skin (rare in darkly-pigmented people)





Early Basal Cell Carcinoma



Early Basal Cell Carcinoma





Basal Cell Carcinoma

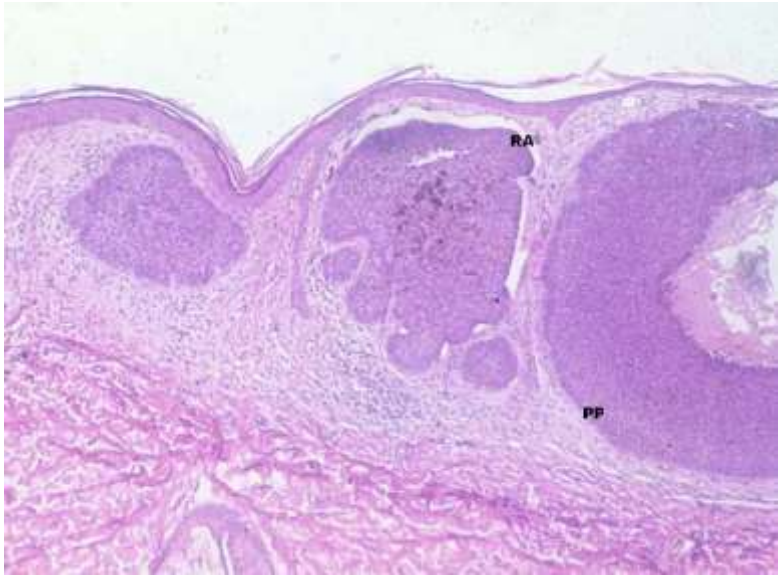


Basal Cell Carcinoma

Objective:

- Location
- Types
  1. Sclerosing
  2. Noduloulcerative



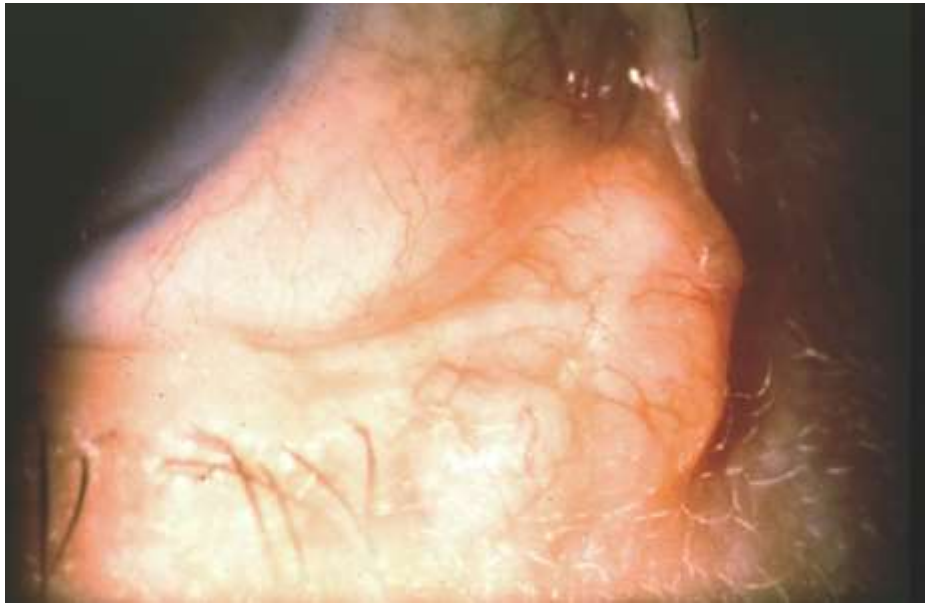


- Early forms look like vascularized nodules
- Varying degrees of central umbilicated ulceration
- ”Pearly”, rolled borders
- Pigmentation
- Surface may become inflamed, infected or both

Ulcerative Basal Cell Carcinoma



Basal Cell Carcinoma
Nodular Ulcerative (left) Sclerosing Type (right)





Basal Cell Carcinoma
Sclerosing Type

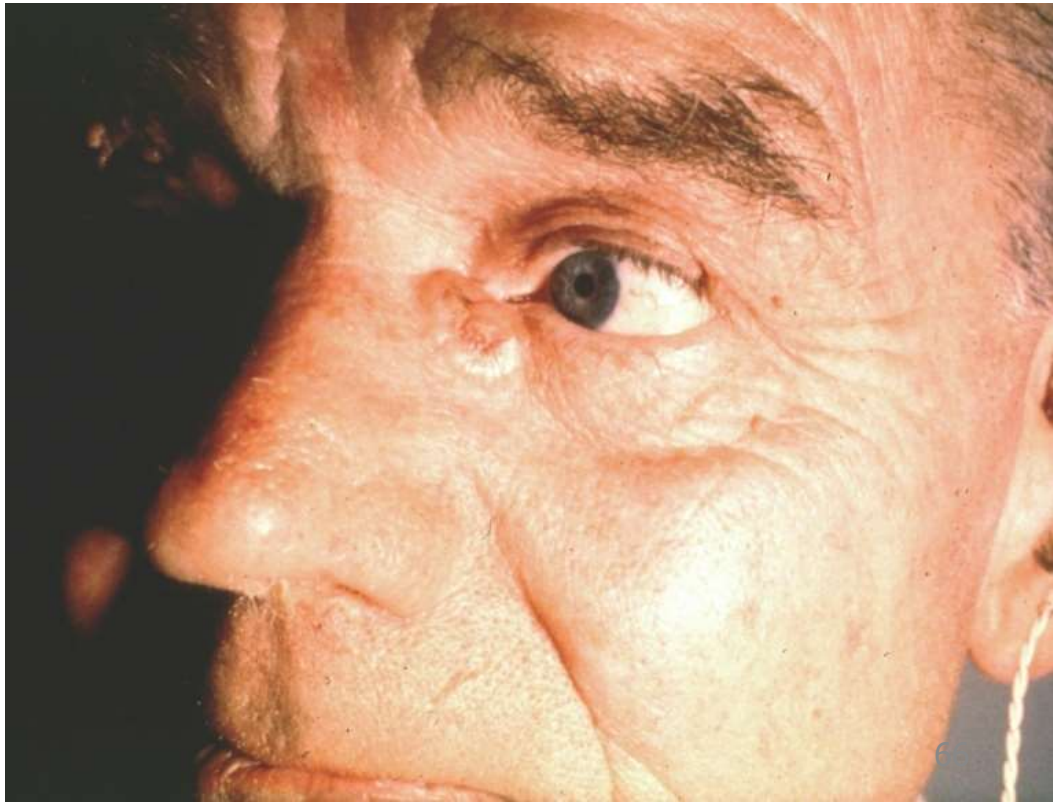


Basal Cell Carcinoma Nodulo-Ulcerative



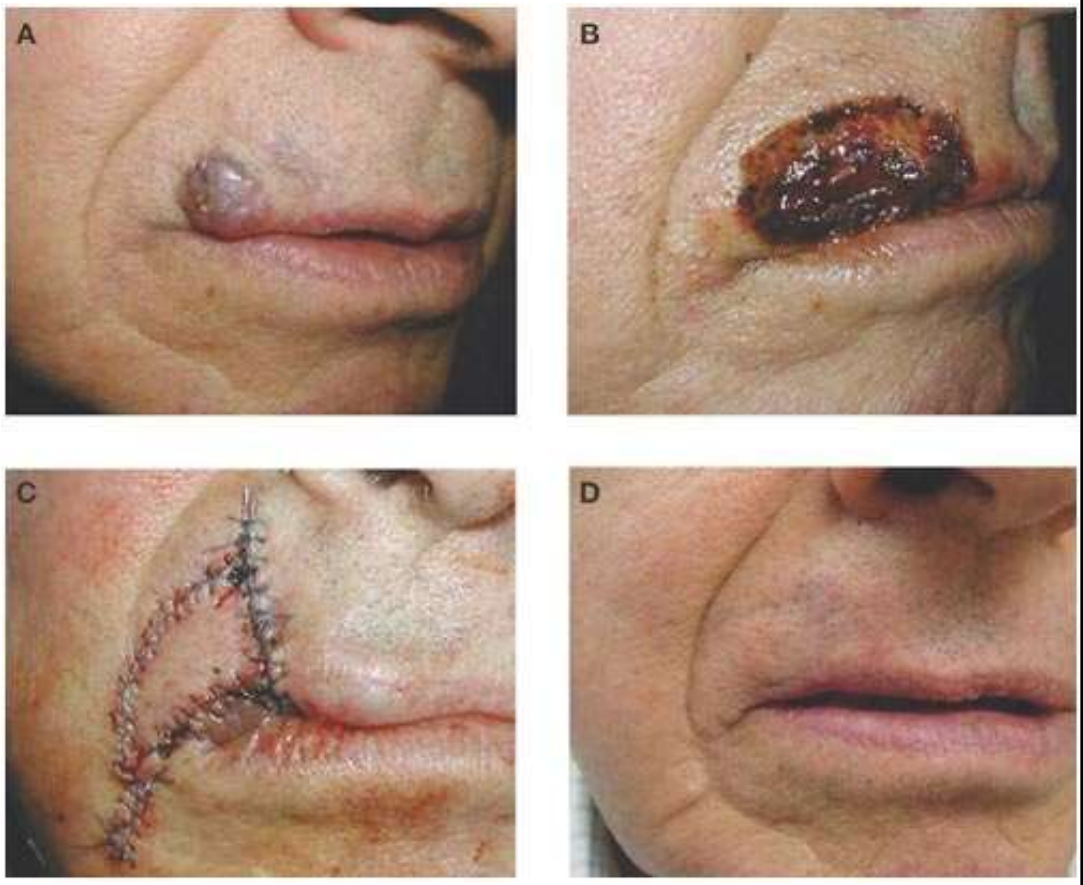
Basal Cell Carcinoma Assessment/DDX

- Metastasis uncommon
- Inner canthal lesions require more rapid attention
- Consider all lid lumps & bumps!
  - Keratoacanthoma
  - HSV lesion

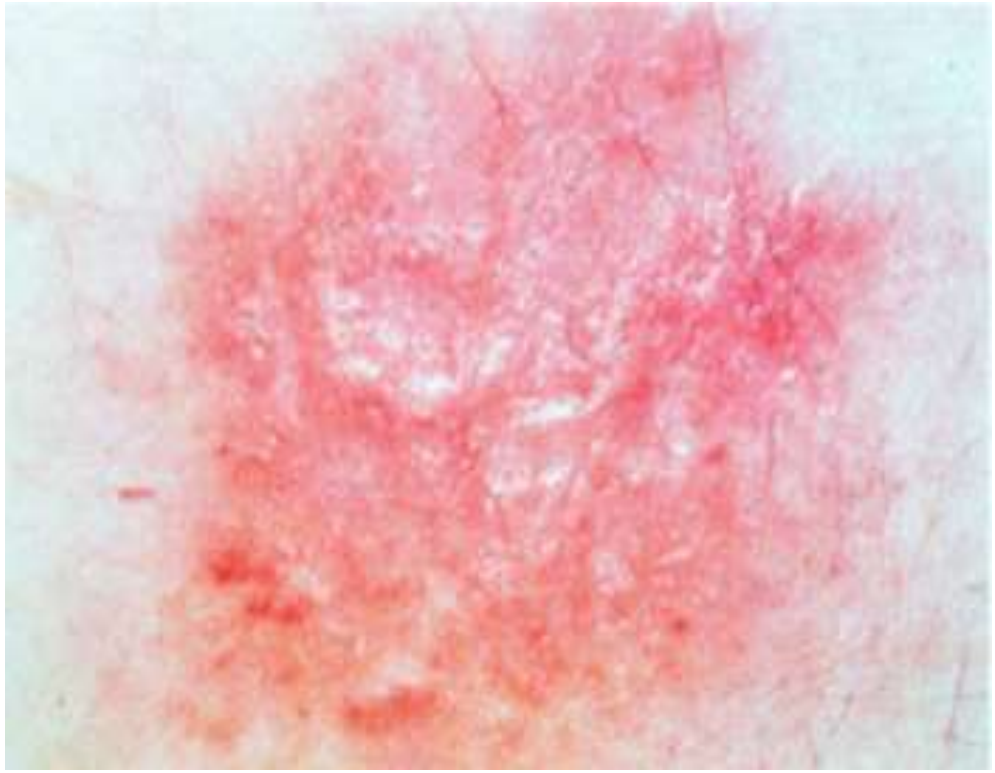


Basal Cell Carcinoma Plan/TX

- If secondarily infected or inflamed, run short course of AB-steroid combo
- Derm. consult
- surgical excision & repair
- radiation
- cryotherapy
- curettage (scoop out)
- electrodessication
- Imiquimod cream (Aldara)
- 5-FU (fluorourocil)
- PDT (photodynamic therapy) (injected in vessel)
- Pt. education



Basal Cell Carcinoma

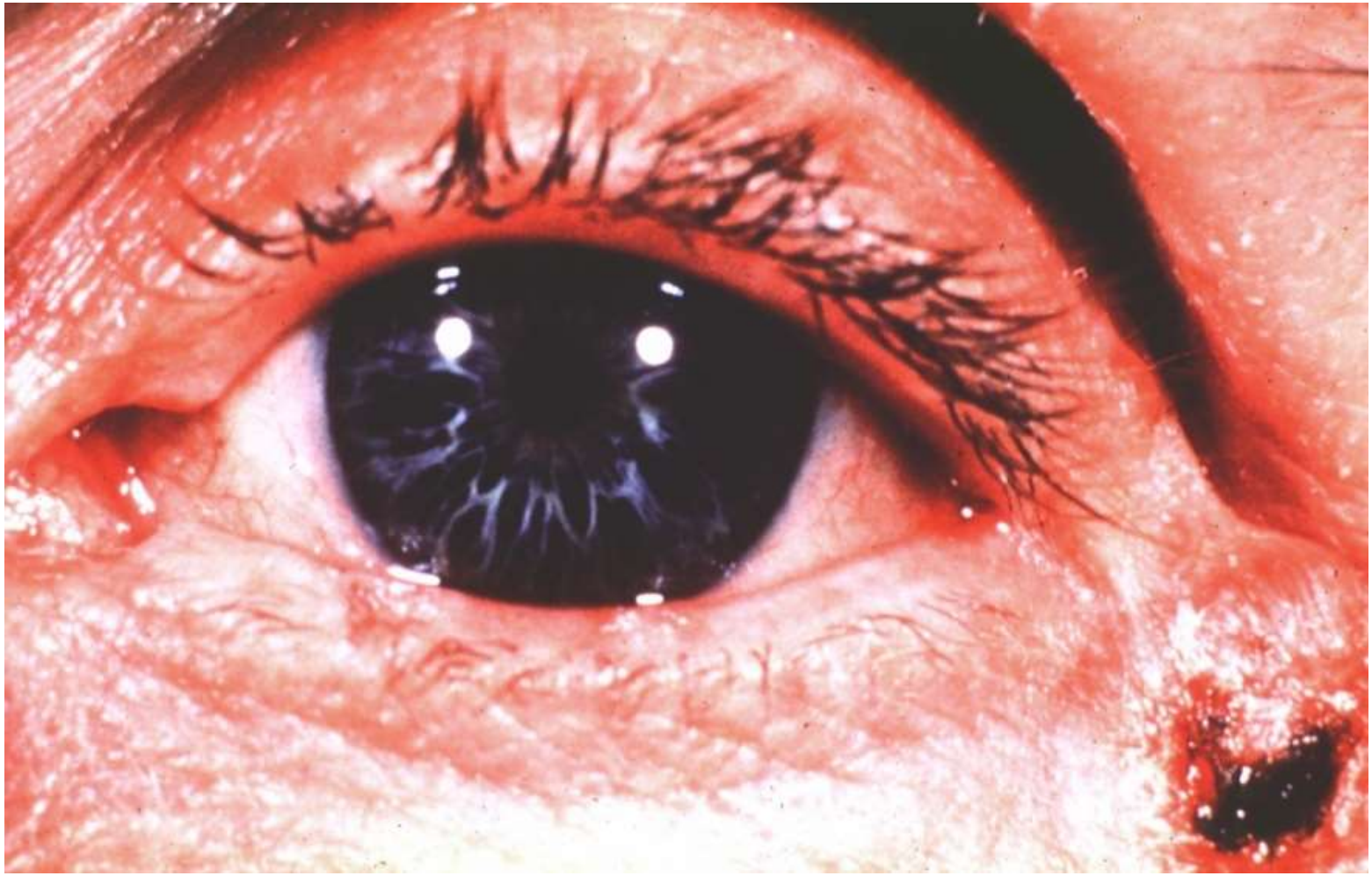




Basal Cell Carcinoma



Basal Cell Carcinoma Infected and Inflamed



Basal Cell Carcinoma Infected and Inflamed



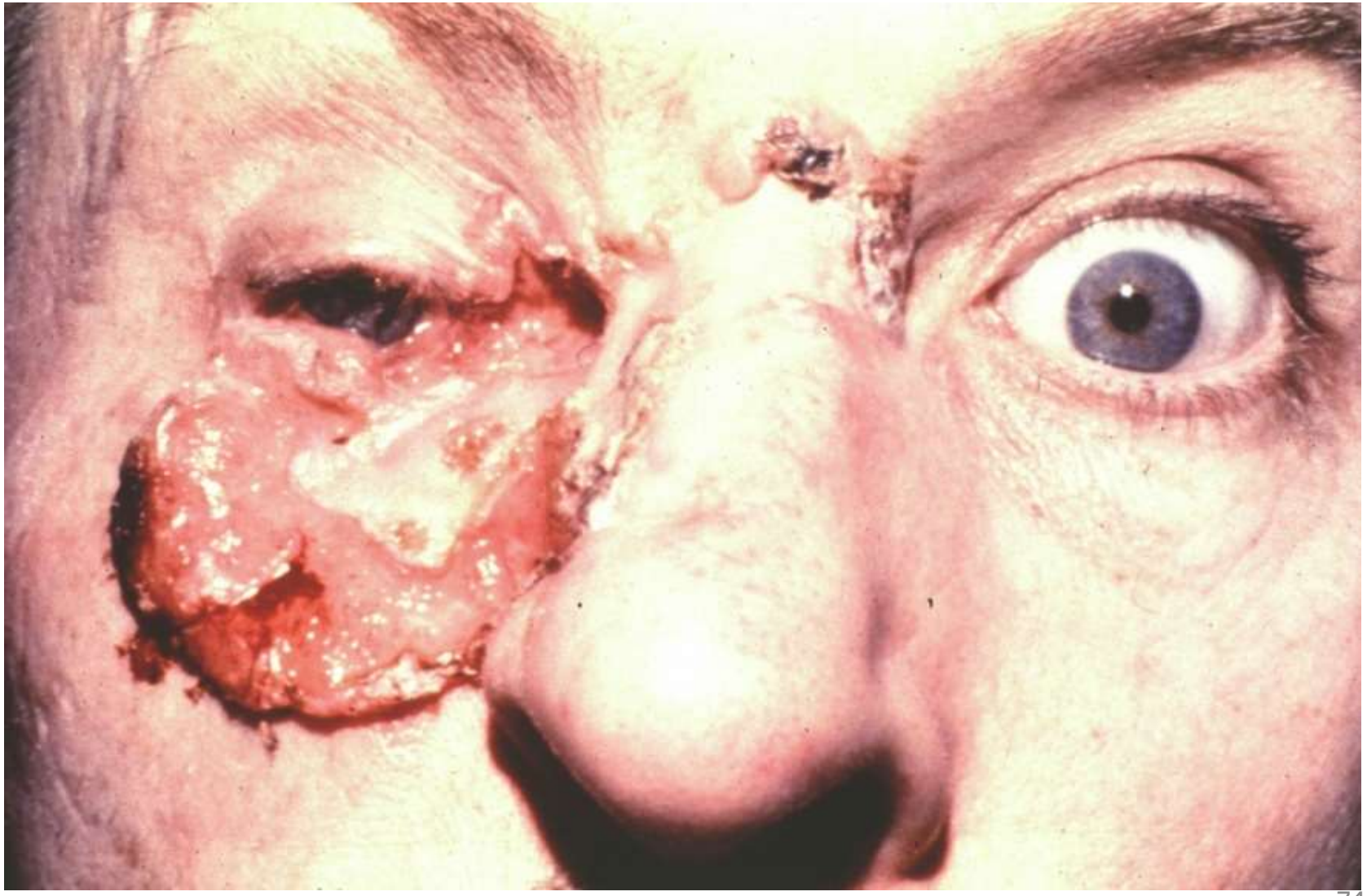
Basal Cell Carcinoma



Basal Cell Carcinoma



Basal Cell Carcinoma



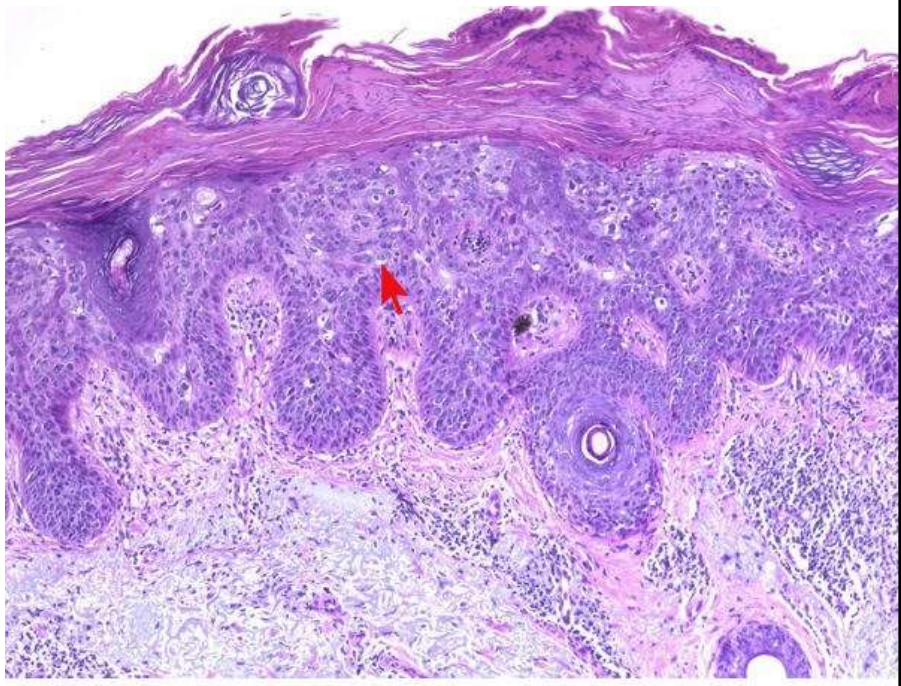
SQUAMOUS CELL CARCINOMA

2th most common tumor

Squamous Cell Carcinoma

- Malignant tumor
- 3RD MOST COMMON OCULAR TUMOR
- 2nd most common skin cancer
- Etiology: Keratinizing epidermal cells





Intraepidermal atypical squamous cells

Squamous Cell Carcinoma

- Subjective – scab that doesn't heal
- Objective - reddish, raised scaly plaque
- Assessment/DDX - R/O all lid lumps & bumps!
  - Keratoacanthoma
  - sebborheic keratoses
  - cutaneous warts
- Plan/TX - excision, radiation, PDT (photo-dynamic therapy)



Squamous Cell Carcinoma



Squamous Cell Carcinoma



Squamous Cell Carcinoma





MEIBOMIAN GLAND CARCINOMA

not as common tumor. Tumor of glands

carcenoma tumor → epithelium

Meibomian Gland Carcinoma

- Mistaken for persistent, recurrent meibomian gland chalazion
- A type of sebaceous gland carcinoma
- Can mimic a chronic eyelid/conjunctival infection
- Plan/TX
  - Excision
  - Radiation



Meibomian Gland Carcinoma



Meibomian Gland Carcinoma



PRECURSORS TO CANCER

can lead to cancer.




Lone Pine Koala Sanctuary,
Brisbane www.koala.net





Actinic Keratosis

can be mistaken for age related

- = solar keratosis = “sun spots”
- Squamous cell dysplasia

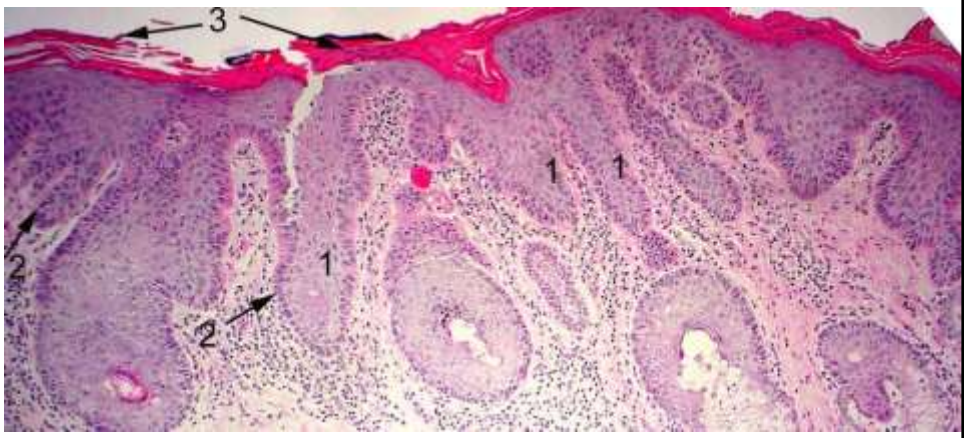
Etiology – UV exposure

Subjective

- Pt unaware
- Pt notices ‘skin spots’

Objective:

- Dry, scaly lesions 2-5 mm
- Minimally elevated





Actinic Keratosis
= sun related

Assessment:

- R/O neoplasia
- R/O keratoacanthoma
- R/O melanoma, nevi, verrucae, papilloma
- R/O seborrheic keratoses



Plan/TX:

- Dermatology consult
- Cryo, cutterage, topical anti-cancer cream (e.g. imiquimod), PDT

Actinic Keratosis



Actinic Keratosis



Actinic Keratosis



Keratoacanthoma

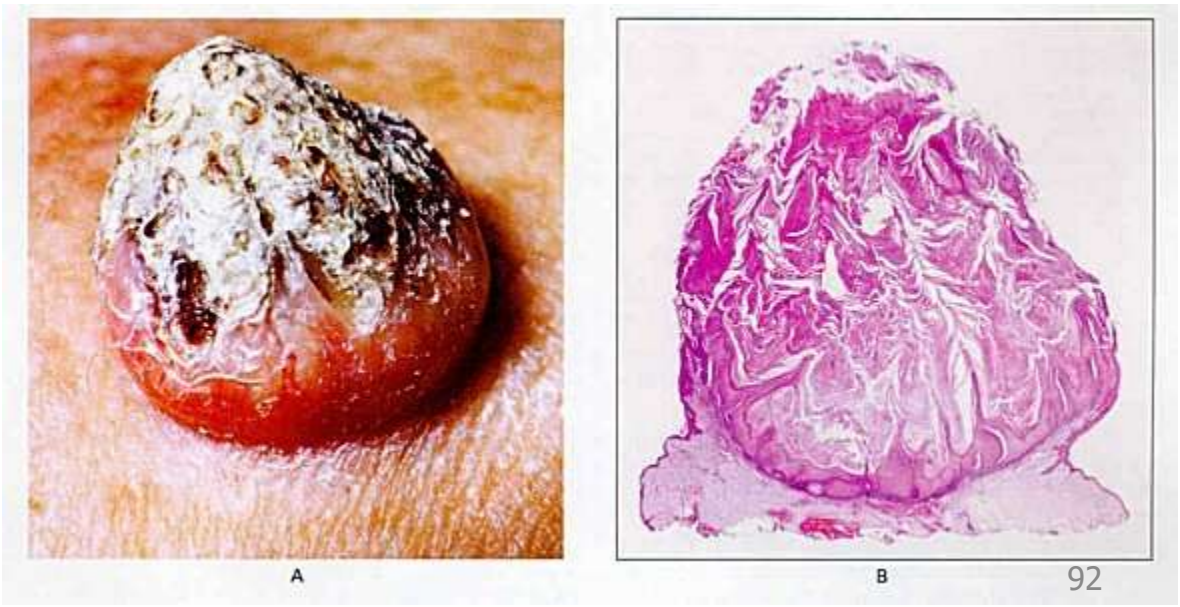
looks like horns, dark center

- Papular lesion originating from sebaceous glands
- Benign proliferation of squamous epidermal cells

Etiology – unknown (possibly UV, trauma, age)

Subjective – fast growing





Keratoacanthoma

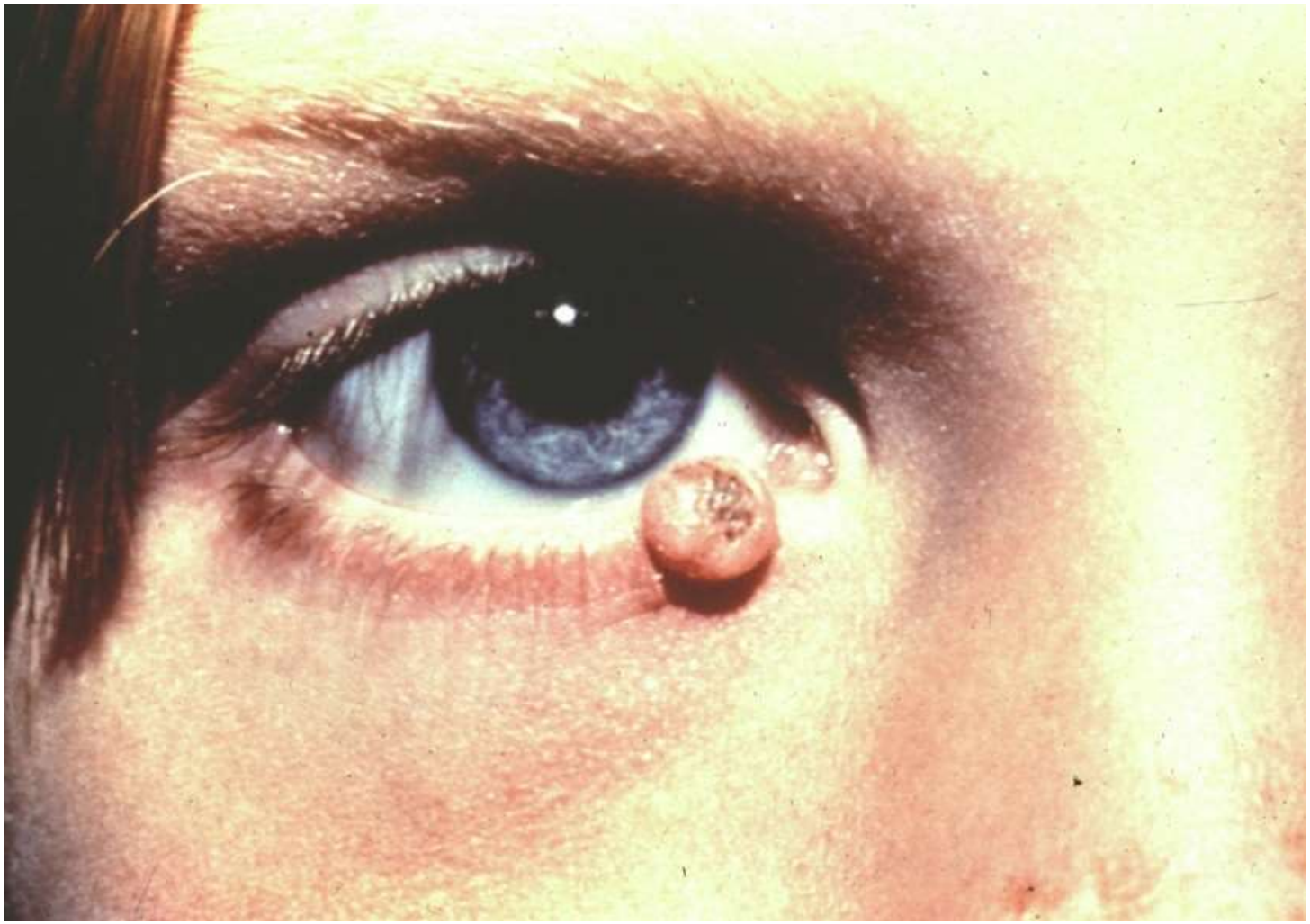
Objective can start as just red spot.

1. Starts as erythematous papule
2. Grows into firm, raised indurated nodule
   - up to 2 cm
3. Keratin filled crater
4. Color
5. Spontaneously regresses



- Assessment - R/O SCC
- Plan/TX - Derm. consult, excision

Keratoacanthoma



Keratoacanthoma



Keratoacanthoma



Keratoacanthoma



Lentigo maligna (Hutchinson's freckle)

- A sub-type of melanoma *in situ*
- Associated with sun exposure (Australia, Hawaii)
- Above age 40; mean age 65. Fair-skin association
- Primarily in head and neck
- Flat pigmented patches
- 15% malignancy risk



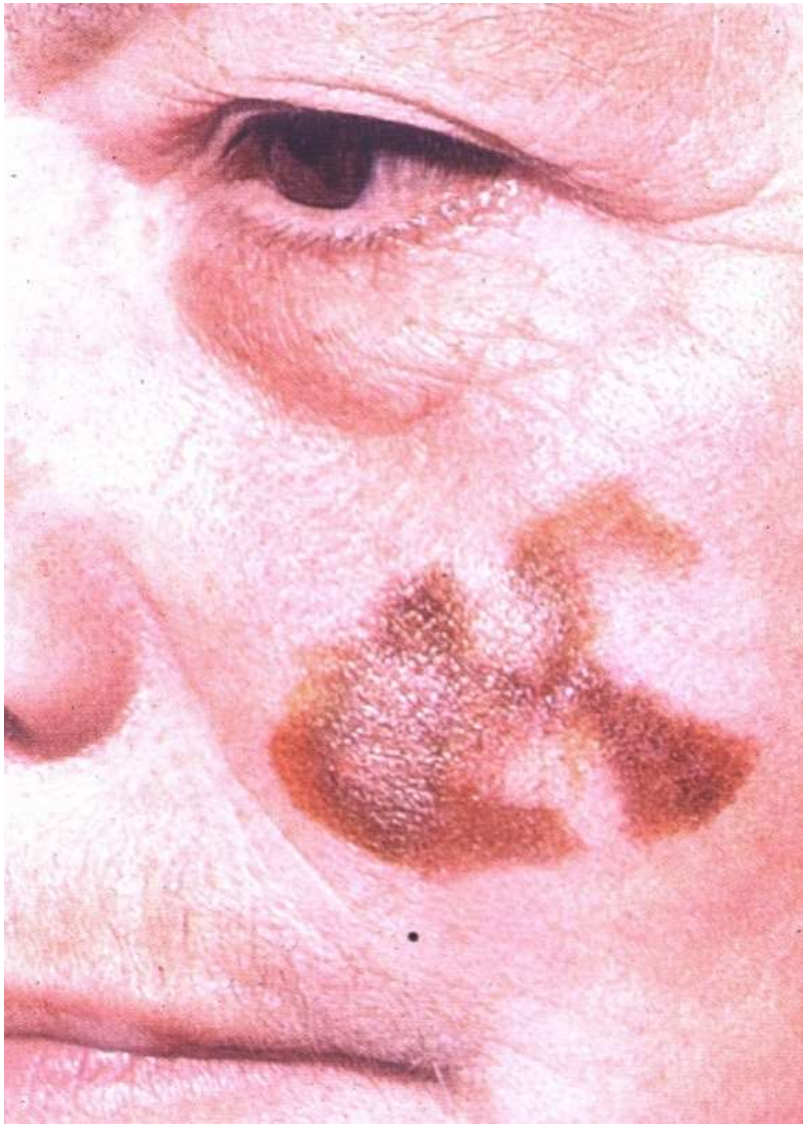


Lentigo maligna (Hutchinson's freckle)

- Objective - dark macular lesion with irregular serpiginous pigment and margins
- Assessment - R/O other pigmented lesions
- Plan - Refer for biopsy – surgical removal



Lentigo maligna (Hutchinson's freckle)



Lentigo maligna (Hutchinson's freckle)



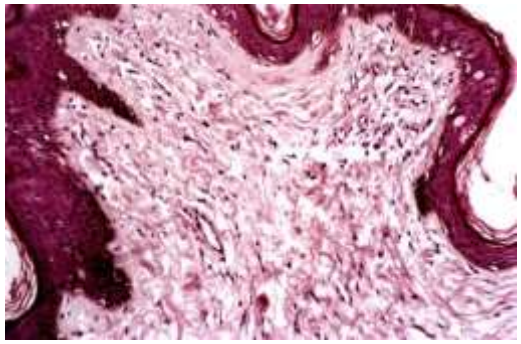
BENIGN SKIN LESIONS

Acrochordon

half of people have this



- = “skin tag”, soft warts
- Incidence 46%
- small, soft, common, benign, usually pedunculated neoplasm found particularly in the obese
- Etiology - Localized paucity of elastic tissue



Mgt:

- Snip it
- Freeze it
- Zap it
- Chromium picolinate 200 mcg BID?

Looseness of skin



Acrochordon (Skin Tag)



- armpit (the most common)
-

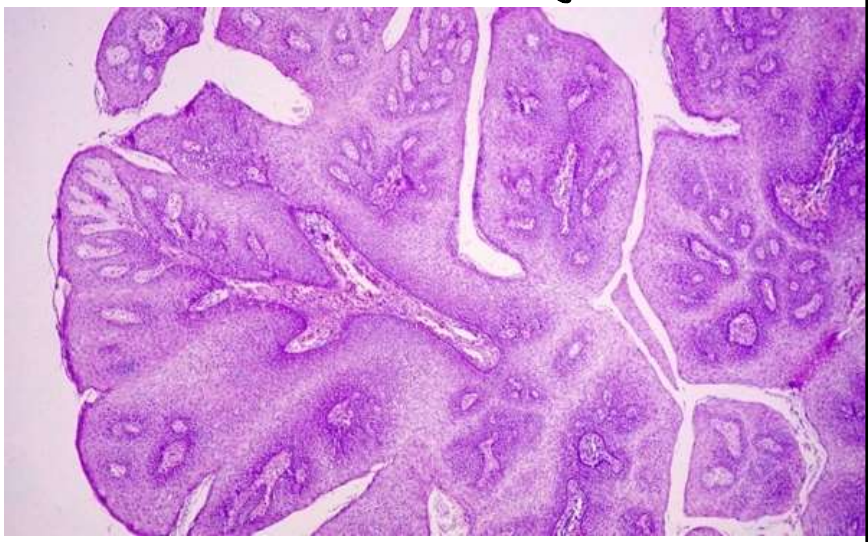
bigger one you put anastetich in.



Papilloma

relatively simple to remove. No sicores because are flat.

- Benign epithelial tumor
- Etiology – epithelial overgrowth
- Lobulated (like a brain) in appearance







Papilloma - Objective

lash loss it is something to worry about.

- Size and shape
- Pigmentation
- Number
- Location
- Avascular
- Growth
- Surface is roughened but not eroded



Papilloma - Assessment

R/O other lumps & bumps:

- Nevus
- Keratoses
- Seborrheic cyst
- Verrucae
- Molluscum
- Squamous cell carcinoma
- Other neoplasia





Papilloma Plan/TX

- Reassurance and pt. education
- Removal
- Derm. consult
- RTC q. 3-6 mos.
- RTC q. 1 yr.





- Papillomae can be sessile (flattened) or pedunculated (elongated)
- They can be the same color as skin or pigmented
- They often look like cauliflower!

Papilloma

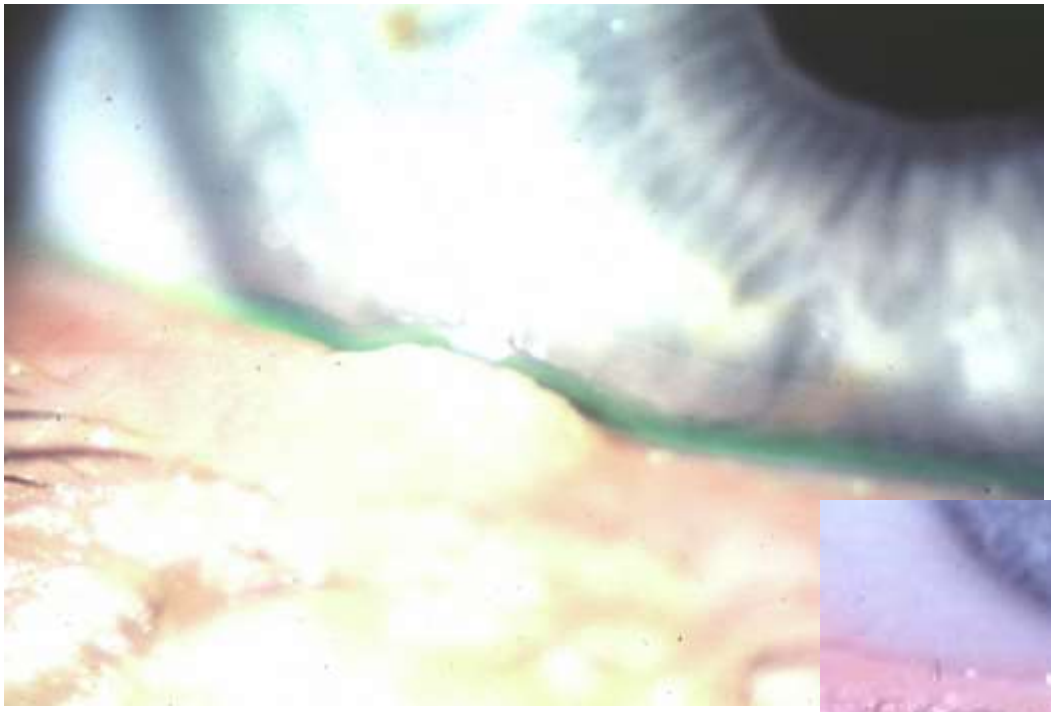


Pigmented Papilloma





Papilloma

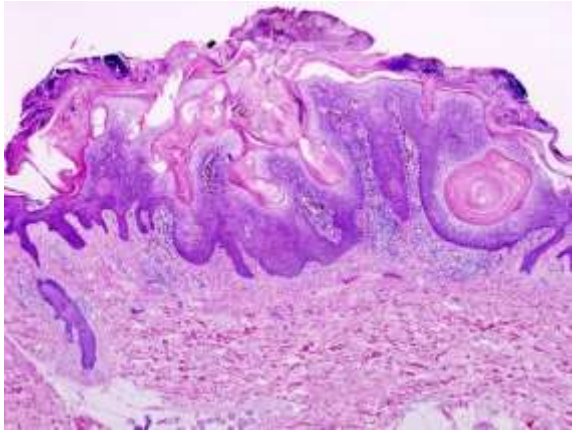




Seborrheic keratosis



- = “barnacles of old age”
- Etiology – basal cell hyperplasia
- Hyperkeratinized plaques
- Most common benign epidermal tumor in the elderly



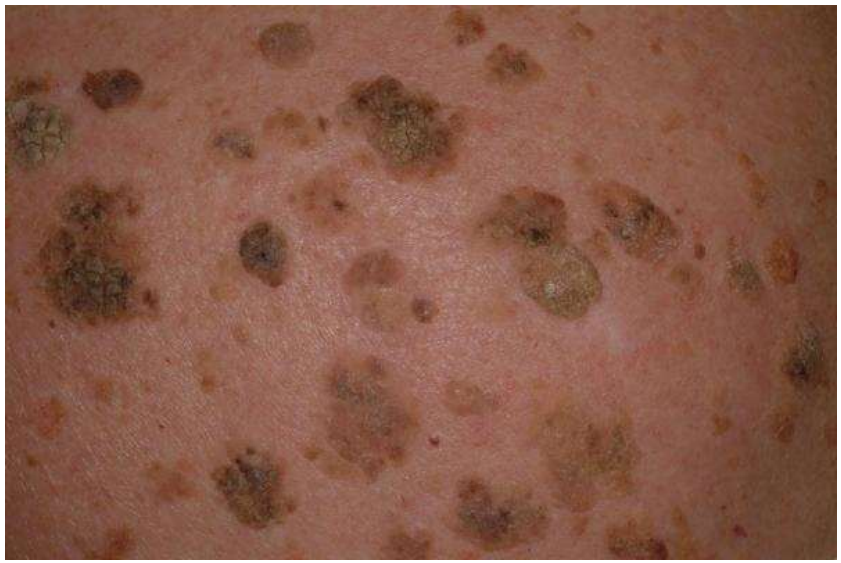




Seborrheic keratosis

Objective:

- Flat or slightly elevated 1-6 mm round to oval
- Waxy, greasy or scaly lesion





Seborrheic Keratosis Assessment

Rule-out:

- neoplasia
- keratoacanthoma
- melanomas, nevi, verrucae, papilloma, actinic keratosis
- Plan/TX
  - Derm. Consult
  - cryo, excision, cuterrage
  - RTC q. 6-12 mos.

Normal Mole	Melanoma	Sign	Characteristic
		Asymmetry	when half of the mole does not match the other half
		Border	when the border (edges) of the mole are ragged or irregular
		Color	when the color of the mole varies throughout
		Diameter	if the mole's diameter is larger than a pencil's eraser

Seborrheic Keratosis



Seborrheic Keratosis



Seborrheic Keratosis



Seborrheic Keratosis



Lichen planus

- Common pruritic eruption affecting pts between 30-60 yo
- Reported in approximately 1% of all new pts seen at dermatology clinics
- Flat topped violaceous papules (2-4 mm)
- Etiology - cell mediated immune response of unknown origin





Atopic / allergic derma. related

Lichen planus

Also commonly affects buccal (mouth) mucosa

Management:

- Biopsy prudent
- Steroids
- Vitamin A
- Retinoids



Lichen planus



Lichen planus





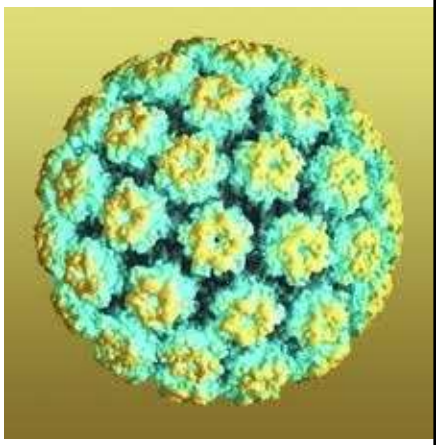
(looks like basal cell)

VIRAL SKIN LESIONS

Verrucae

- = viral warts
- Etiology - Virus (HPV – Human Papilloma Virus)
- Plantar wart (verruca plantaris) – wart on sole or toes of foot
- Benign skin tumor

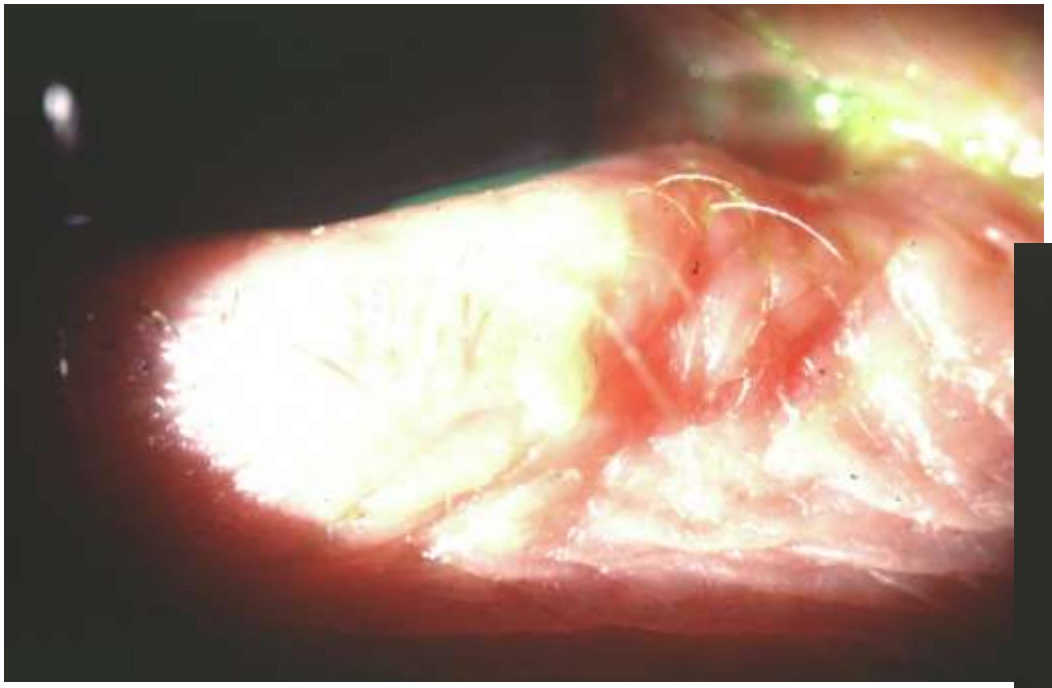
often on the fingers
also in the mouth

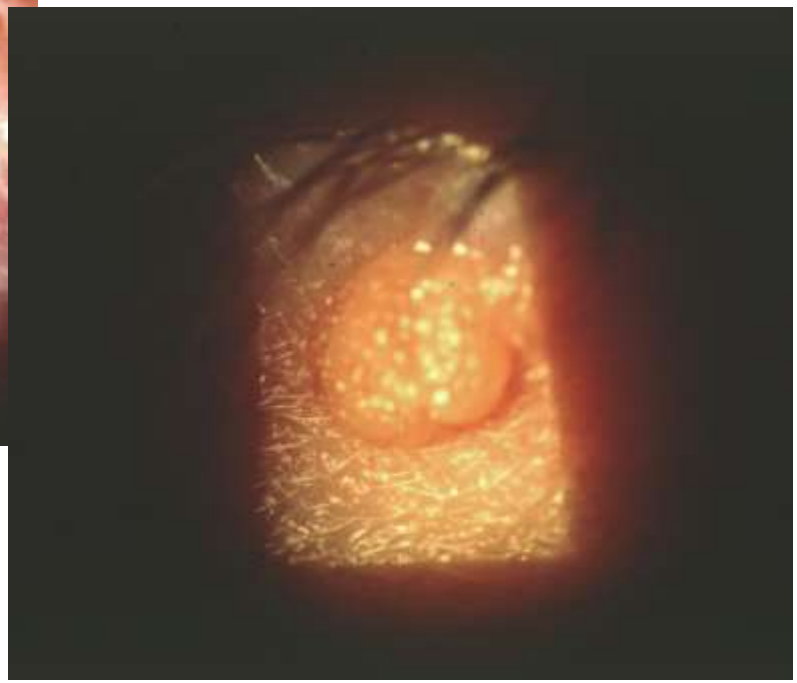






Verrucae





Verrucae





Verrucae

Objective

- Single or multiple non-secreting warts
- Color
- Various shapes:
  1. Verruca plana
  2. Verruca vulgaris
  3. Verruca digitata

Assessment/DDX

- R/O other lumps & bumps – papilloma, molluscum contagiosum

Verrucae Plana





Verrucae Vulgaris





Verrucae Digitata

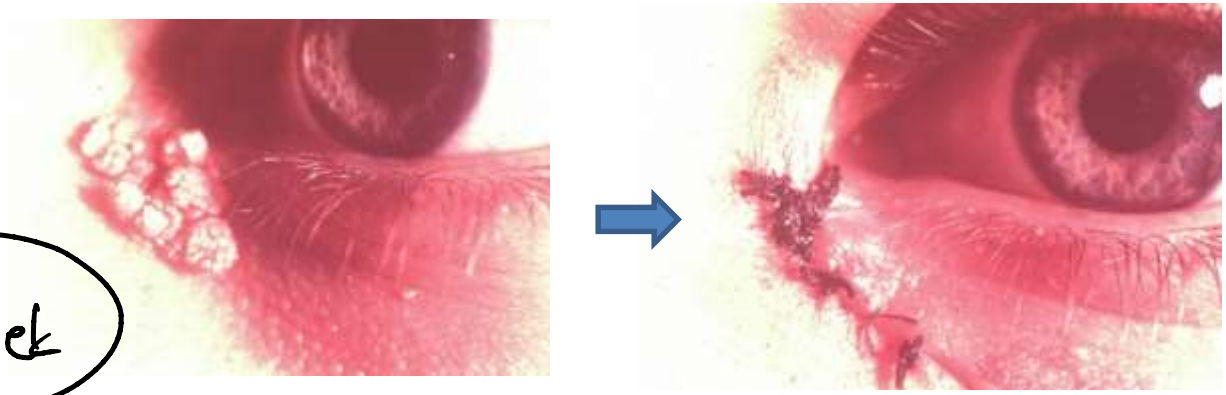


Verrucae Plan/TX

are contagious (nalezljivi)

- Resolution
- Reassurance and patient education
  - Contagious on skin contact, dressing room floors
- Removal
  - Excision
  - Keratolytic agents (dichloroacetic acid)
  - Laser
  - Cryo
  - Duct tape?

leave it for a week



Multiple verruccae removed with excision

Verrucae



Verrucae



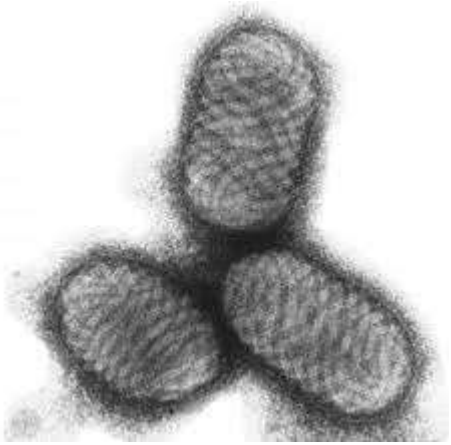


Molluscum contagiosum

one of the most if not the most contagios "diccsa"

- Benign skin tumor
- Etiology - Virus (pox), association with STD and HIV
- Contagious (skin-skin, swimming pools, etc.)
- Children can pass it between them frequently

perfectly round





Molluscum contagiosum

- Almost exclusively human disease
- Member of the Poxvirus family; DNA virus
- Four main types (I-IV)
  - Type I: over 95% of infections
  - Type II: in 60% of HIV infections
  - Types III and IV: very rare
- Inoculation via skin transmission
  - Person to person
  - Autoinoculation (here it on the arm and then rub the chin and you have it on the face)

Molluscum contagiosum

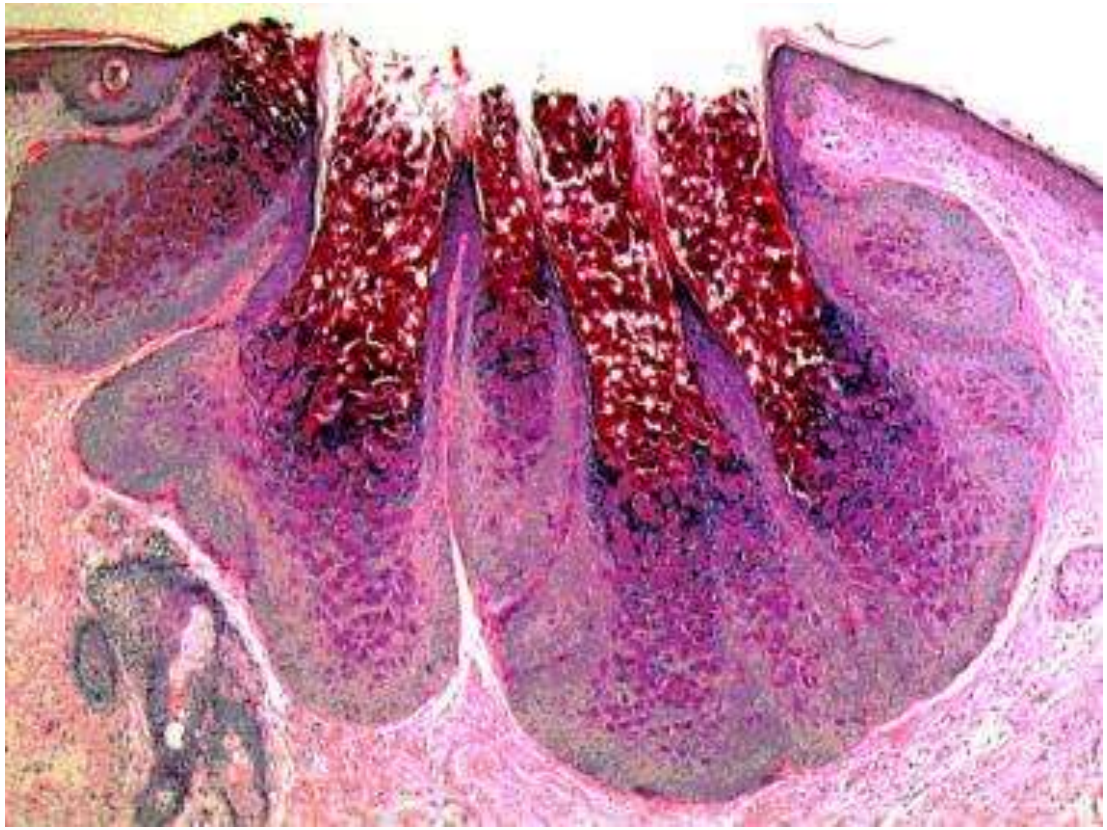
Objective:

- Umbilicated nodule with central core
- Elevated, round, waxy, pearly
- Follicular conjunctivitis
- Multiple
- Keratitis





Lesion crush/squash preparations can confirm Henderson-Paterson bodies (intracytoplasmic virions)



Viral incubation and replication leads to epithelial cellular proliferation with umbilicated core



Molluscum contagiosum



Assessment:

- DDX – R/O BCC

Plan/TX:

- If quiet, leave alone
- If central discharge, express
- Excision, cryo, topical acids, oral medication
- Home remedies (variable benefit)
- Pt. ed. – cleanliness, transmission
- If large number of lesions (100+), consider HIV

Molluscum contagiosum

Topical treatments:

- Podophyllotoxin cream (0.5%)
  - Fetus toxicity; avoid with pregnancy
- Iodine
- Salicylic acid (ASA, Aspirin)
- Potassium hydroxide
- Tretinoin cream
- Cantharidin (blistering agent)
- Imiquimod (T cell modifier)
  - not recommended for children due to possible adverse events and not effective in children



Oral treatment:

- Cimetidine



HOME REMEDIES FOR
MOLLUSCUM CONTAGIOSUM



Apple Cider Vinegar



Tea Tree Oil



Hydrogen Peroxide



Indian Lilac



Australian Lemon Myrtle



Grapefruit Seed Extract



Colloidal Oatmeal



Clove Oil



Oil of Oregano



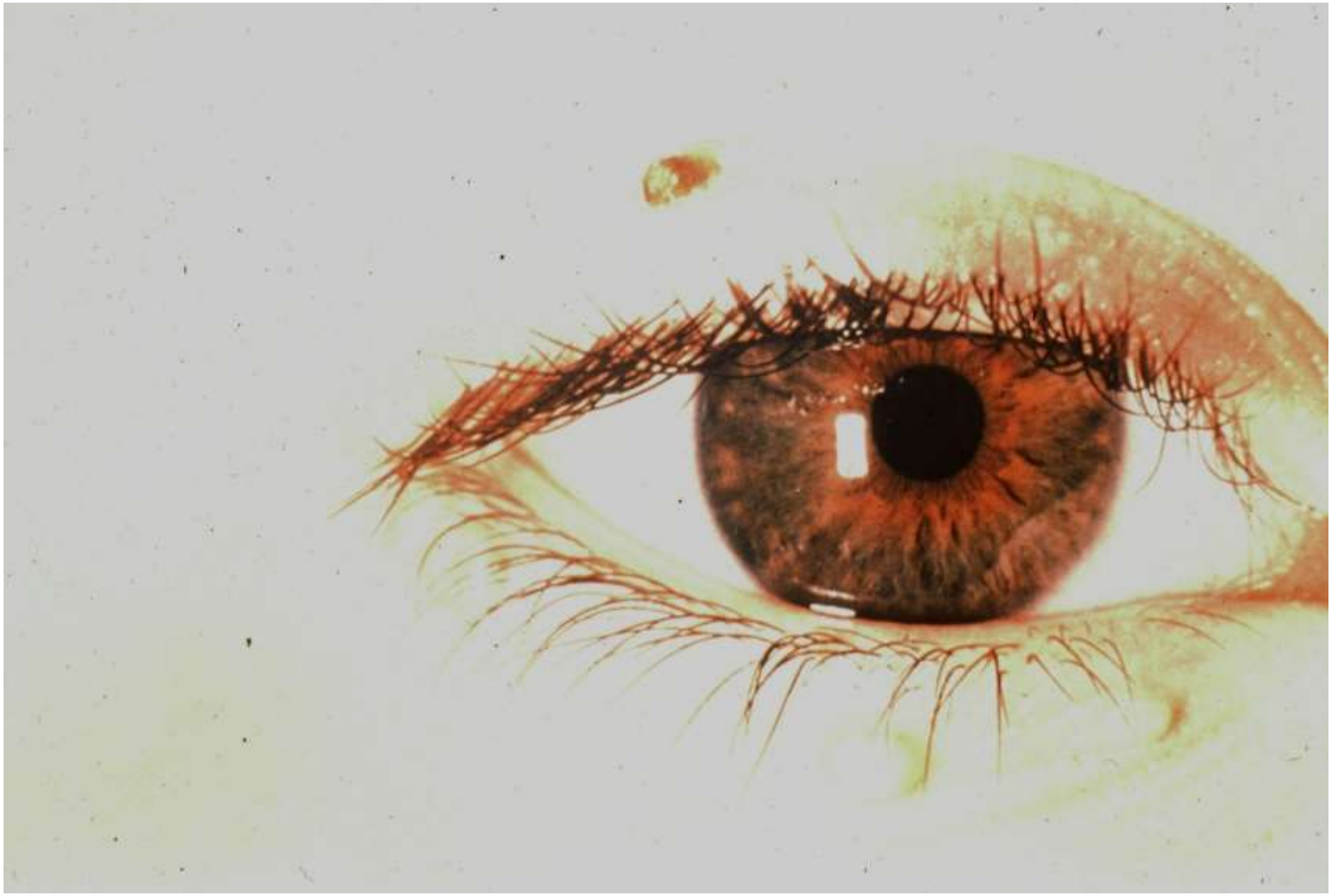
Top 10 Home Remedies

To explore more, visit
www.Top10HomeRemedies.com

Molluscum contagiosum

- Excellent prognosis; tend to self-resolve (months)
- About a third of patients may get recurrence
- Hand wash, cover if possible, avoid scratching
- Avoid physical contact and sharing towels/clothes
- Adolescents/adults: sexual transmission common

Molluscum contagiosum



Molluscum contagiosum



Molluscum contagiosum





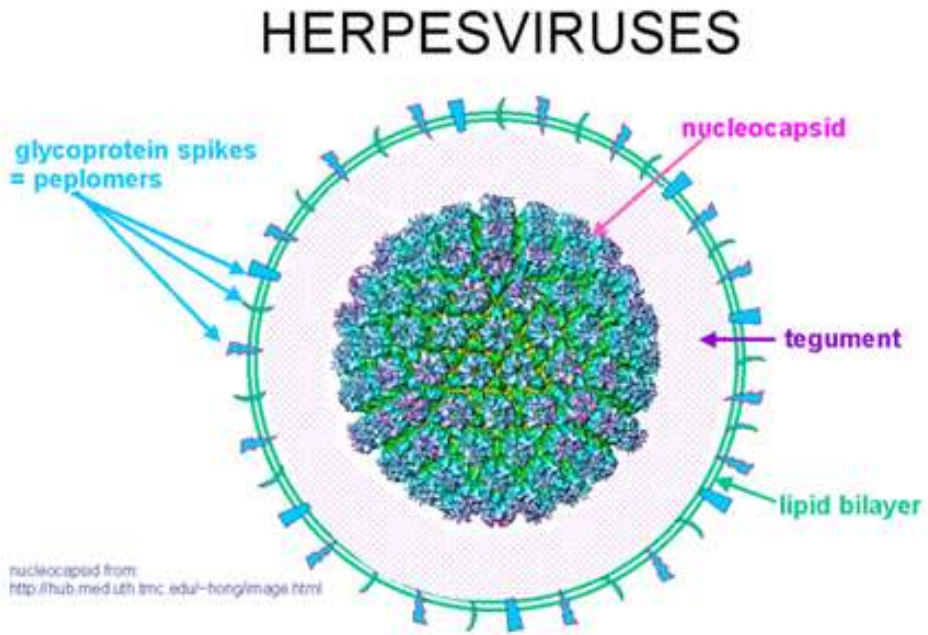
Molluscum contagiosum



Herpetic Eyelid Lesions

There are 2 main herpes eye infections:

- Herpes zoster
- Herpes simplex
- Each of these important virions can cause MANY ocular diseases (including eyelid lesions)



- These will be covered in more detail later in the course

Herpes simplex virus (HSV) primary lesions



Varicella zoster virus (VZV)
aka herpes zoster
Shingles lesions



Hutchinson's sign (nose)



Thoracic dermatome

Shingles

THANK YOU!